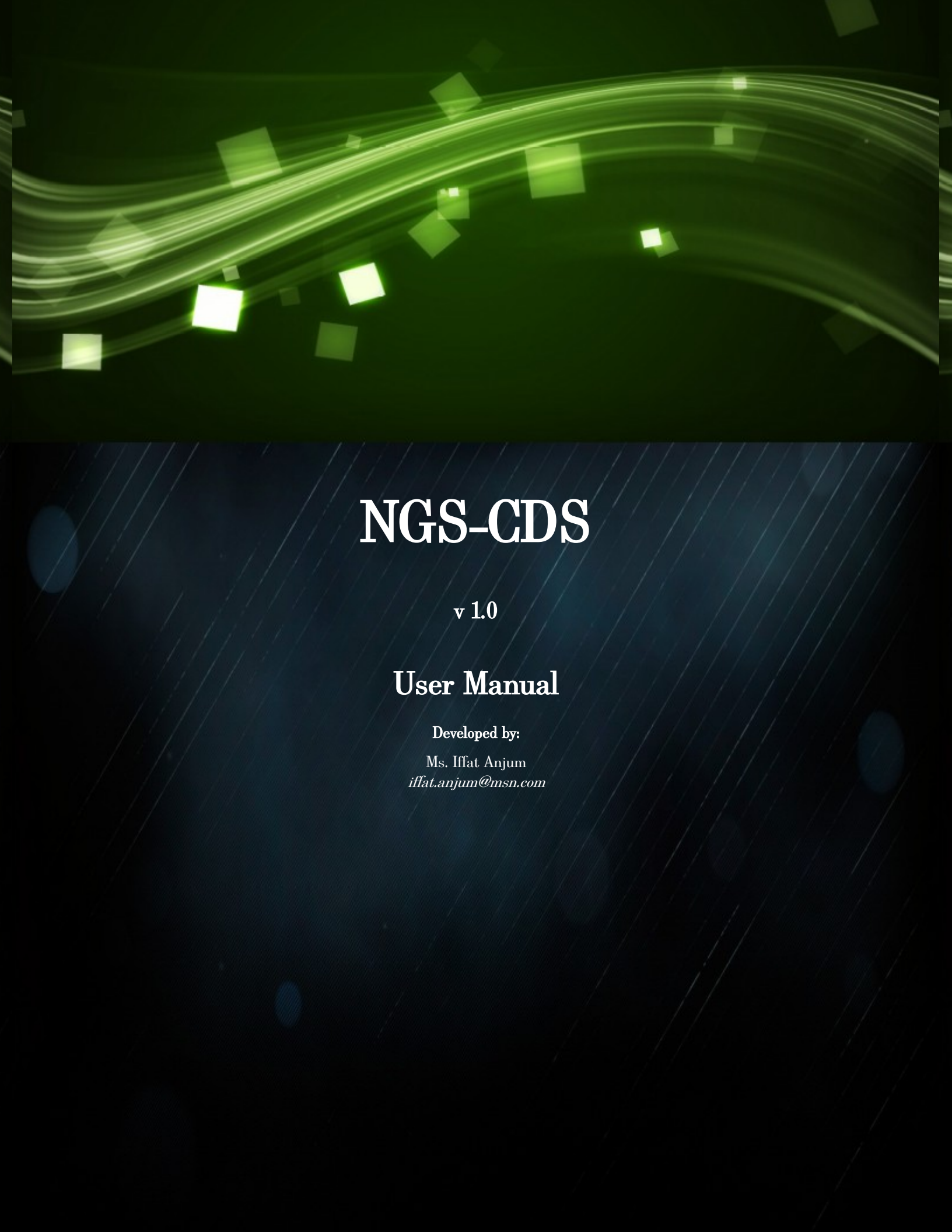




NGS-CDS

v 1.0

User Manual

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Welcome to NGS Cancer Diagnostic Suite - Your Gateway to Streamlined NGS Analysis

1 Introduction:

NGS Cancer diagnostic suite is an open-source software suite designed specifically for **linux** users to revolutionize NGS data analysis in cancer diagnostics. This comprehensive solution features a user-friendly, fully automated GUI, eliminating the need for complex coding. It directly offers a streamlined and efficient workflow for researchers and clinicians. NGS-CDS is typically structured as a collection of modules. Each module specializes in handling a specific type of data format, streamlining the analysis process for researchers. It provides **two** main, open-source pipelines i-e, **RNA-seq analysis pipeline & Variant Calling pipeline** specifically designed for cancer research. These pipelines automate key steps, providing researchers with a streamlined approach to unlocking the secrets hidden within cancer genomes. Additionally, the suite features dedicated GUI interfaces for all the necessary popular tools like **STAR**, **Picard**, and **GATK** within it allowing researchers to leverage their preferred functionalities alongside the core pipelines (Figure 1).

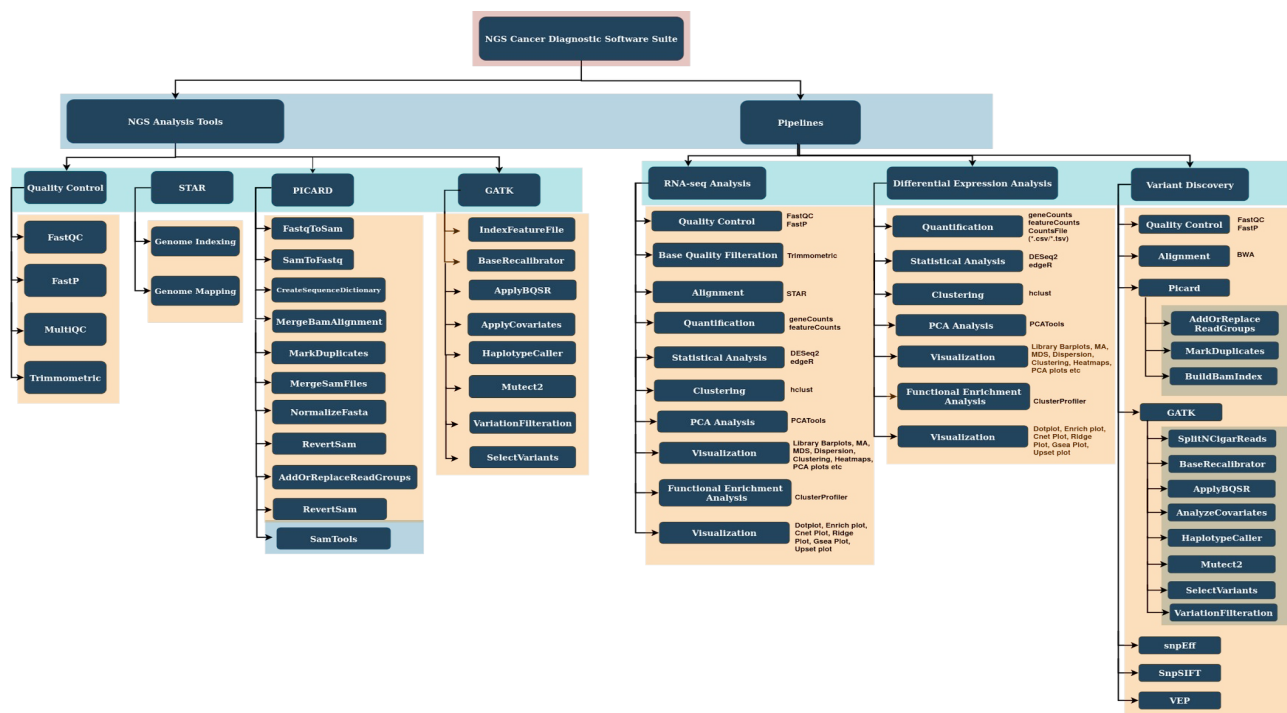


Figure 1: Workflow of NGS Cancer Diagnostic Suite

2 Settings and System Requirements:

Type	Specifications
Operating System(s)	- Linux - Windows with a Linux virtual machine (e.g., VirtualBox, WSL2, or VMware) 64 bit (<i>Recommended</i>) or 32 bit version
Memory	Minimum 8 GB (16 GB recommended for large datasets)
Processor	Quad-core processor or higher recommended
Software Dependencies:	R: Version 4.5.1 Python: Version 3.8 or higher Java Development Kit (JDK): Version 11 or higher

3 Installation:

Launch the installer directly and click '**Begin Installation**' button to start installation or either open the NGS-CDS application interface, go to **File > Setup Wizard** to show installation dialog and click '**Begin Installation**' (Figure 2(a),2(b)).

The installer can also be launched directly from the command line using:

java -jar installNGS.jar

Note: The installation process needs to be performed only once on each system.

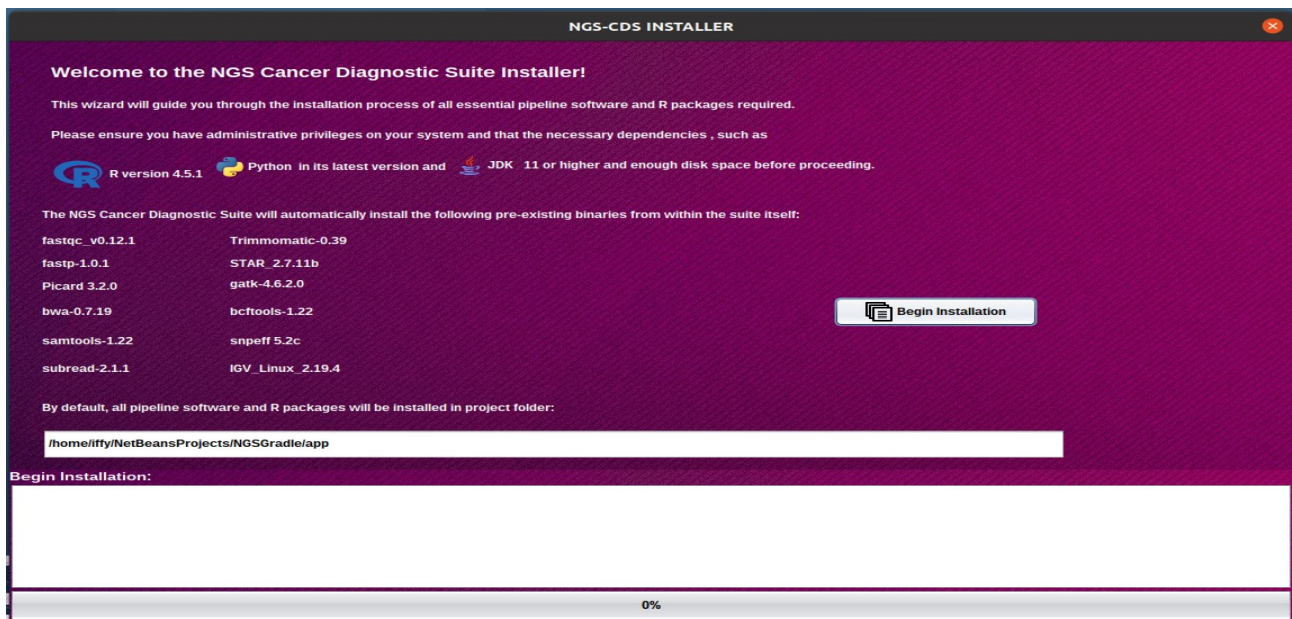


Figure 2(a): NGS-CDS Installation Dialog – Initial Installation Interface

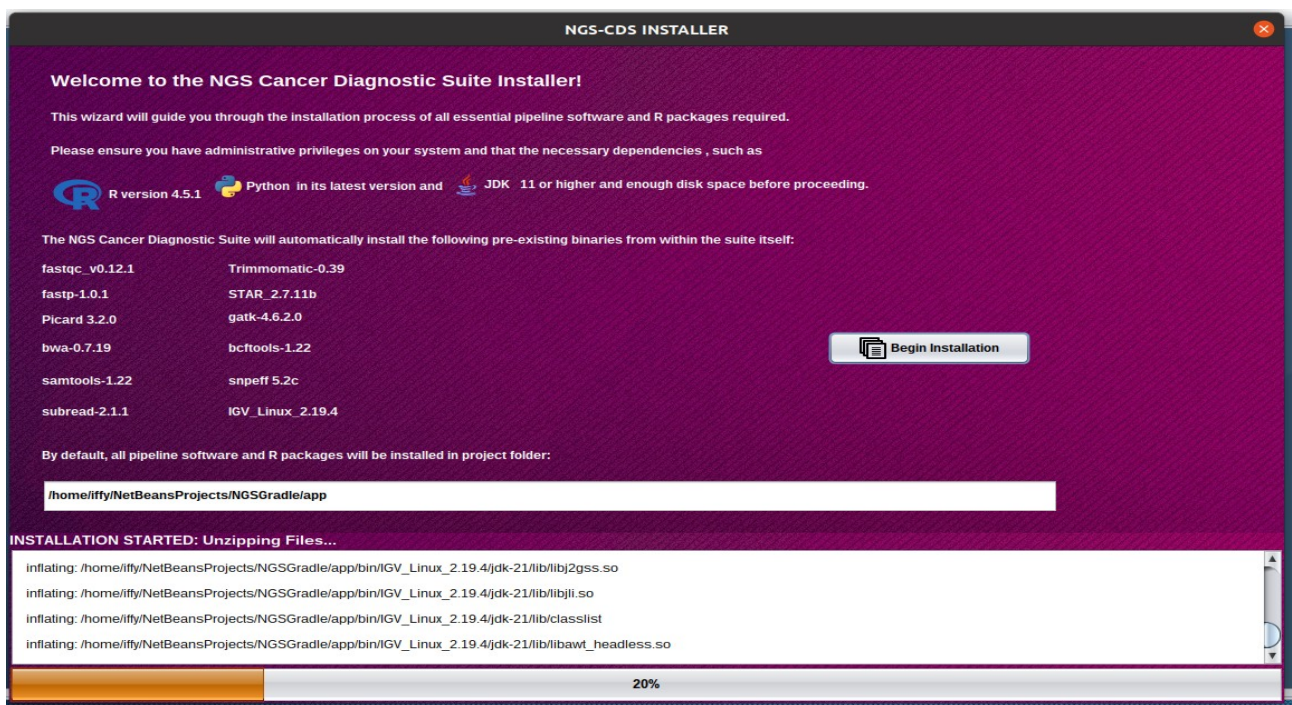


Figure 2(b): NGS-CDS Installation Dialog – Initial Progress Begins

During installtion process, the software may prompt you to enter your (**root**) **administrative password**. This is necessary to installed system required libraries that may be required to R packages system-wide and to set up environment variables and dependencies (Figure 2(c),2(d)).

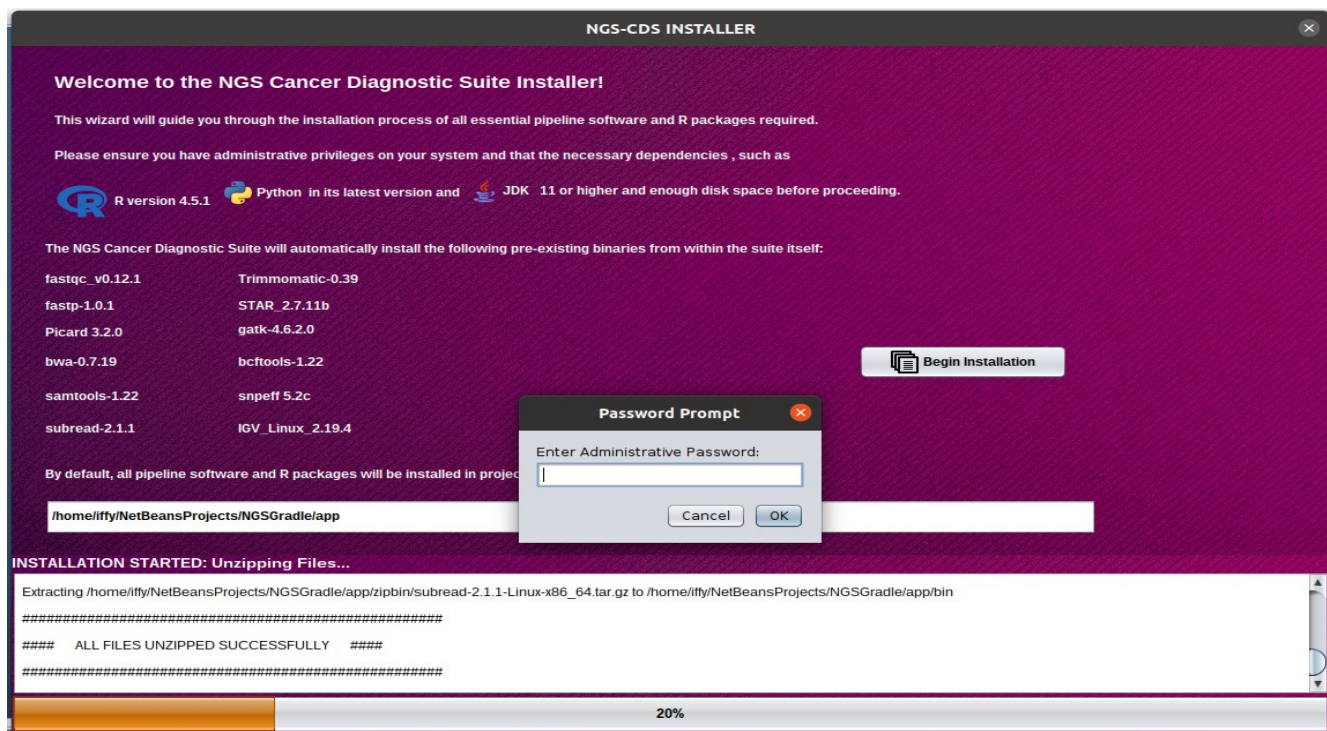


Figure 2(c): NGS-CDS Installation Dialog – Prompt for Administrator Password

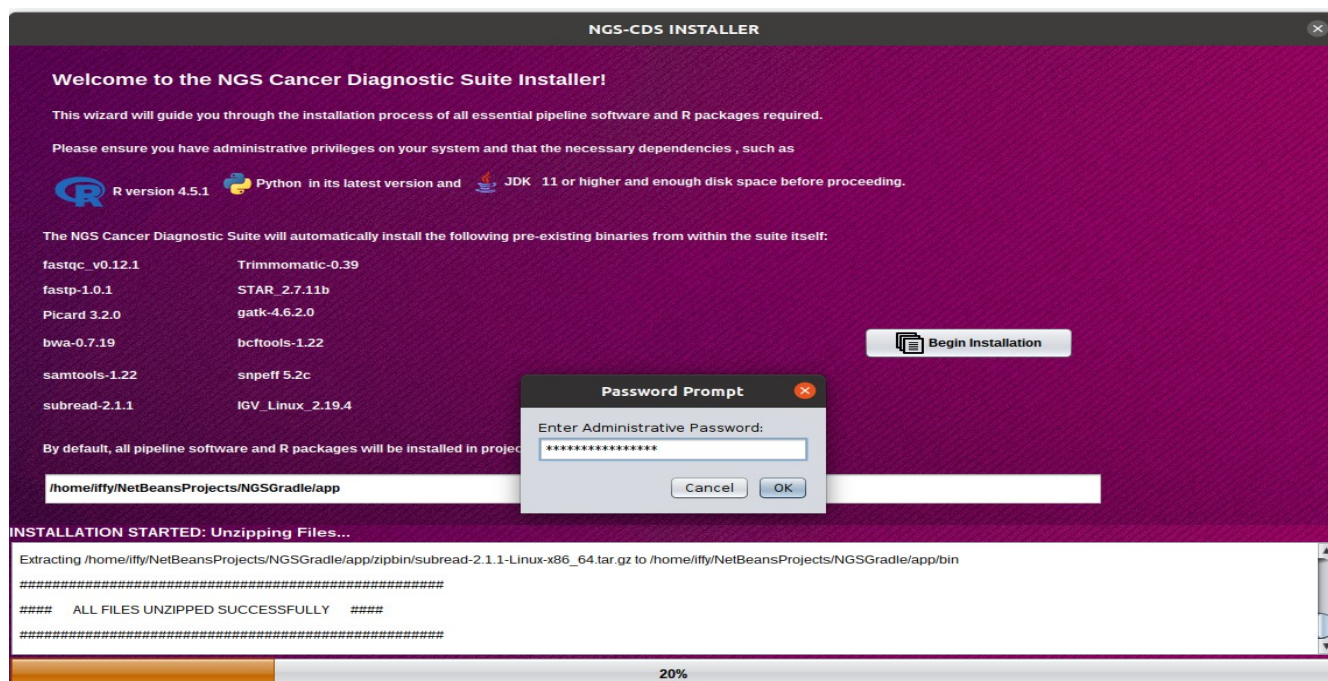


Figure 2(d): NGS-CDS Installation Dialog – Password Entry (Masked as Asterisks)

When prompted enter your sudo/root password. This password is required only onetime. The password will nither be visible nor be stored anywhere (Figure 2(f)).

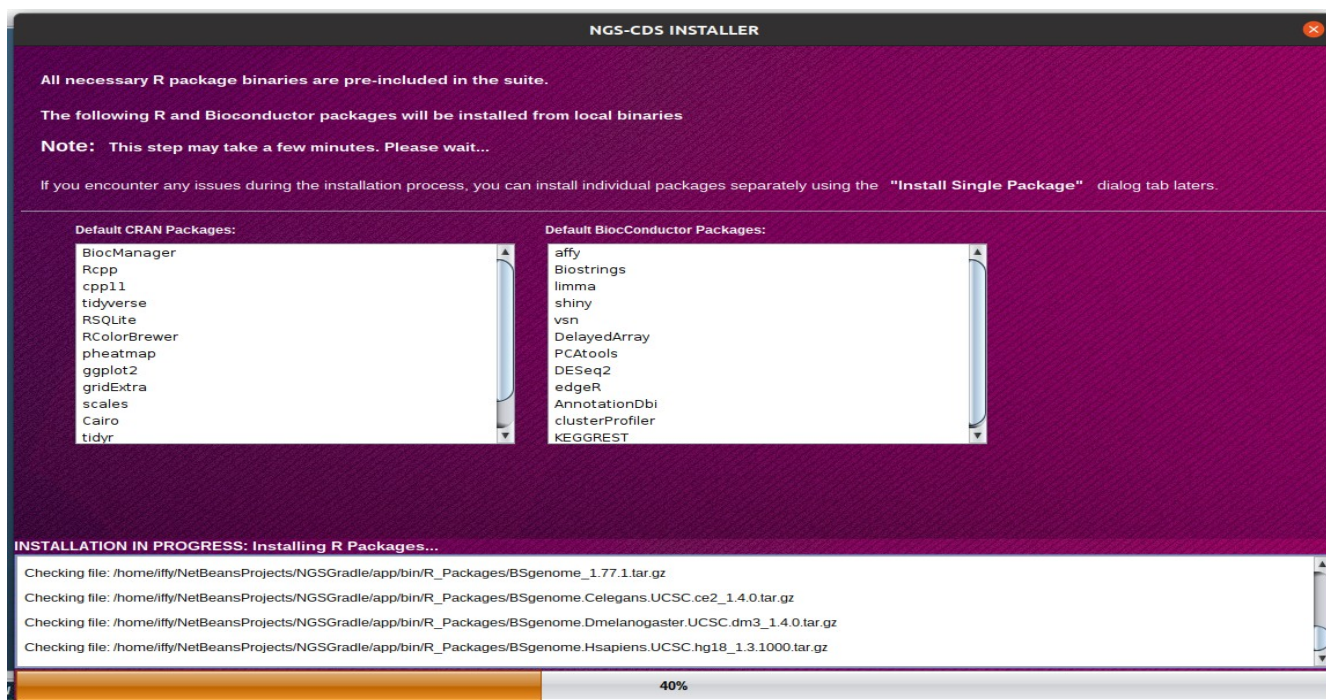


Figure 2(e): NGS-CDS Installation Dialog – R Package Installation in Progress

Once password is enter correctly the installation will continue automatically.

Once the installation is complete, click **FINISH** button and begin using the application `java -jar NGSSuite.jar`

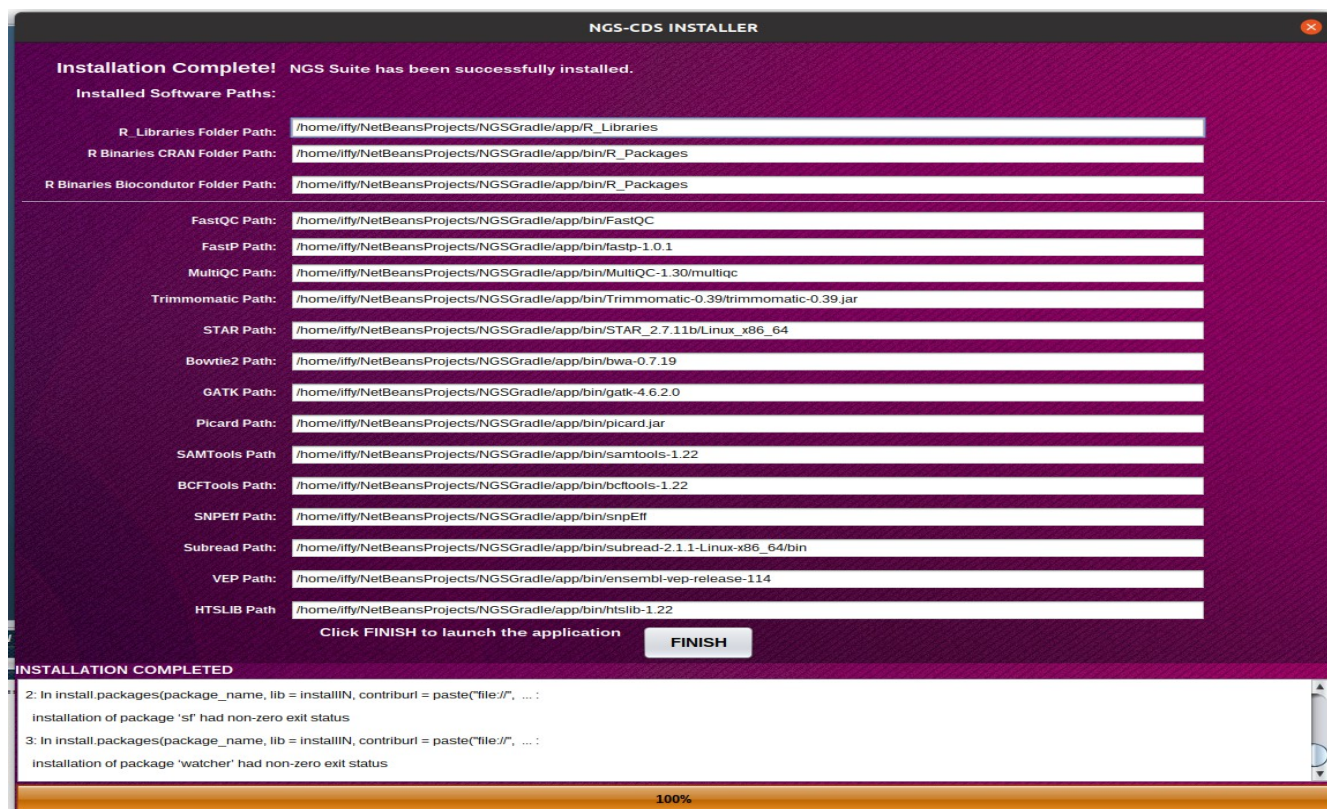


Figure 2(f): NGS-CDS Installation Dialog – Installation Completed Successfully

Handling R Package Installation Errors :

If any R package installation fails during setup, you can manually install the missing packages from within the application: **File > R Packages > Install R Package**

If any package(s) is missing, user can install that package any time either directly from the default binaries folder for CRAN or Bioconductor packages or directly from the internet (Figure 3).

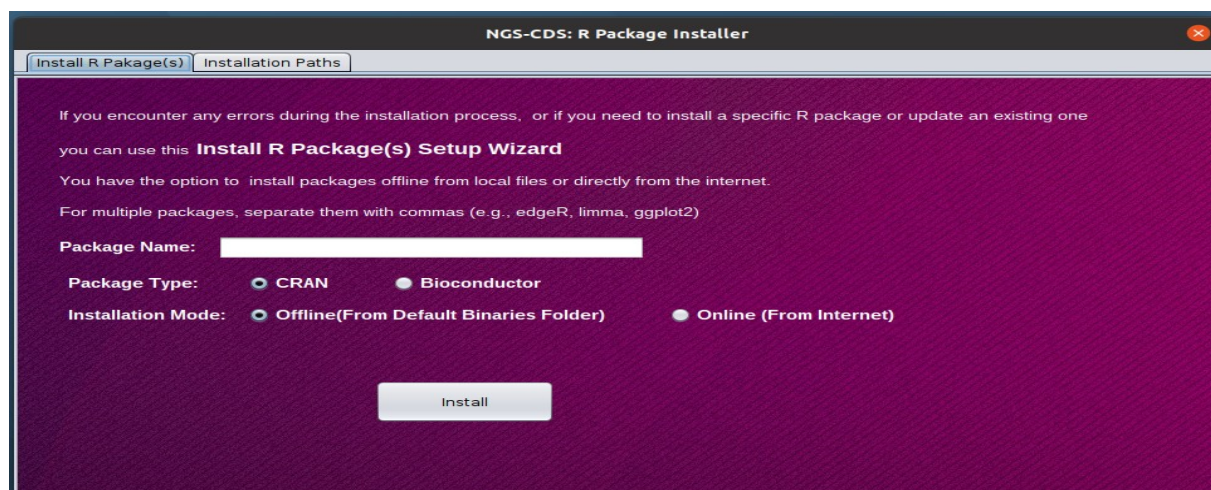


Figure 3: NGS-CDS Manual R Package Installation – Dialog for Installing R Packages Manually in Case of Errors

Two installation modes are provided to the user. **Offline mode** will install libraries from the default pre downloaded binaries and is fast as compared to the other mode and is **recommended**. **Online mode** will download all libraries directly from the internet and install them.

4 Getting Started with NSG Cancer Diagnostic Suite:

If a launcher/shortcut was created during installation, simply **double-click the launcher icon** to open the application (Figure 4(a),4(b)). Or Open a terminal window and run the following command from the installation directory:

java -jar NGSSuite.jar

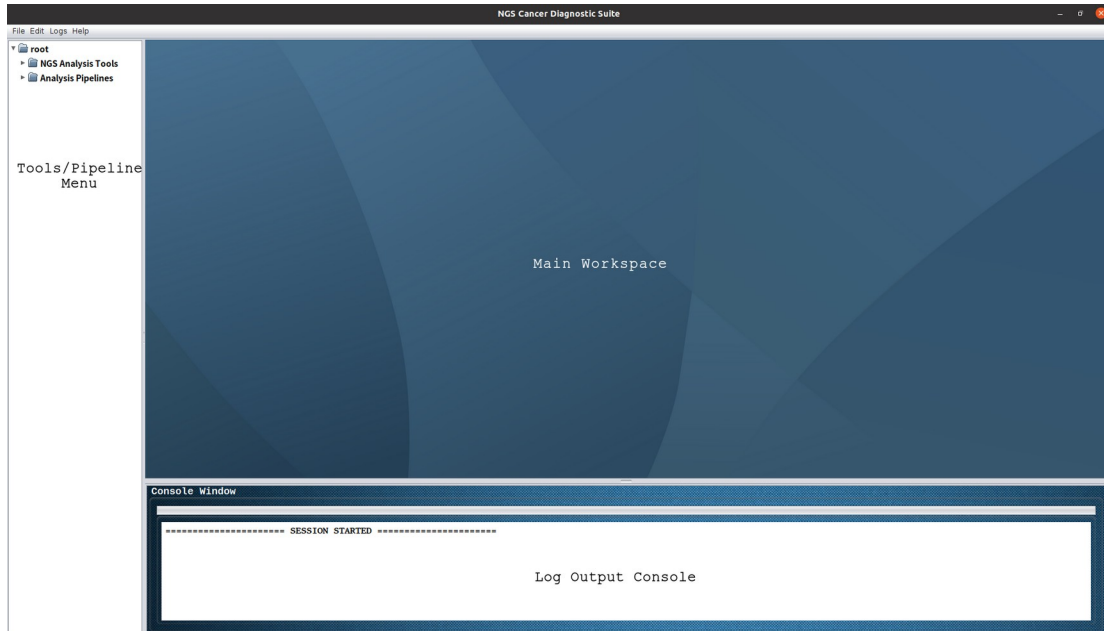


Figure 4(a): NGS Cancer Diagnostic – Main Application Interface

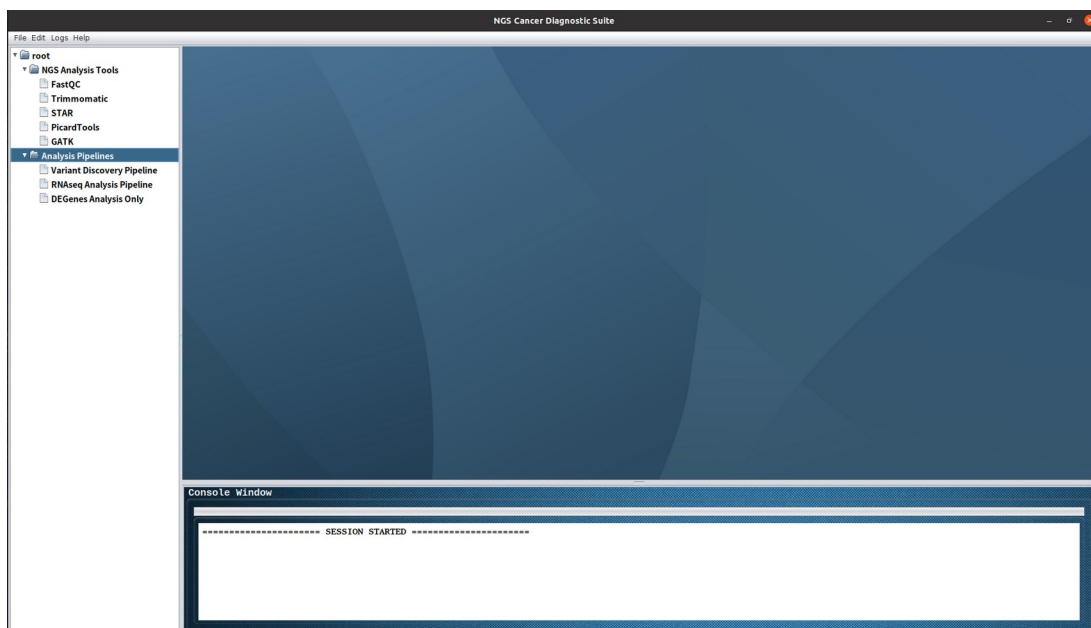


Figure 4(b): NGS Cancer Diagnostic – Main Interface with Sidebar Displaying Pipelines and Tools

5 Running Pipelines:

5.1 RNA-Seq Analysis Pipeline:

RNA Seq is a widely used methodology for the study of transcriptome based on NGS that enables to investigate differential gene expression. The correct identification of differentially expressed genes (DEGs) between specific conditions is a key in the understanding phenotypic variation. This pipeline implements a complete graphical user workflow starting from reading fastq reads to calculating featurecounts, performing DEG analysis using DeSeq and EdgeR and finally performing gene set enrichment analysis (Figure 5).

Step 1: Setting up Output Directory and Input files:

This interface allows users to configure the initial settings for running RNA-seq analysis pipeline. User can specify:

- Output directory,
- Input either single-end or paired-end fastq files,
- Choose the quality control method or even can skip it.
- Choose differential expression method DESeq2, edgeR or even Both.

User can set up any folder as output directory by clicking and browsing to that folder. All the generated results will be stored in that output directory. Its recommended that folder is empty otherwise files might be overwritten.

NOTE: Path should not contain any spaces.

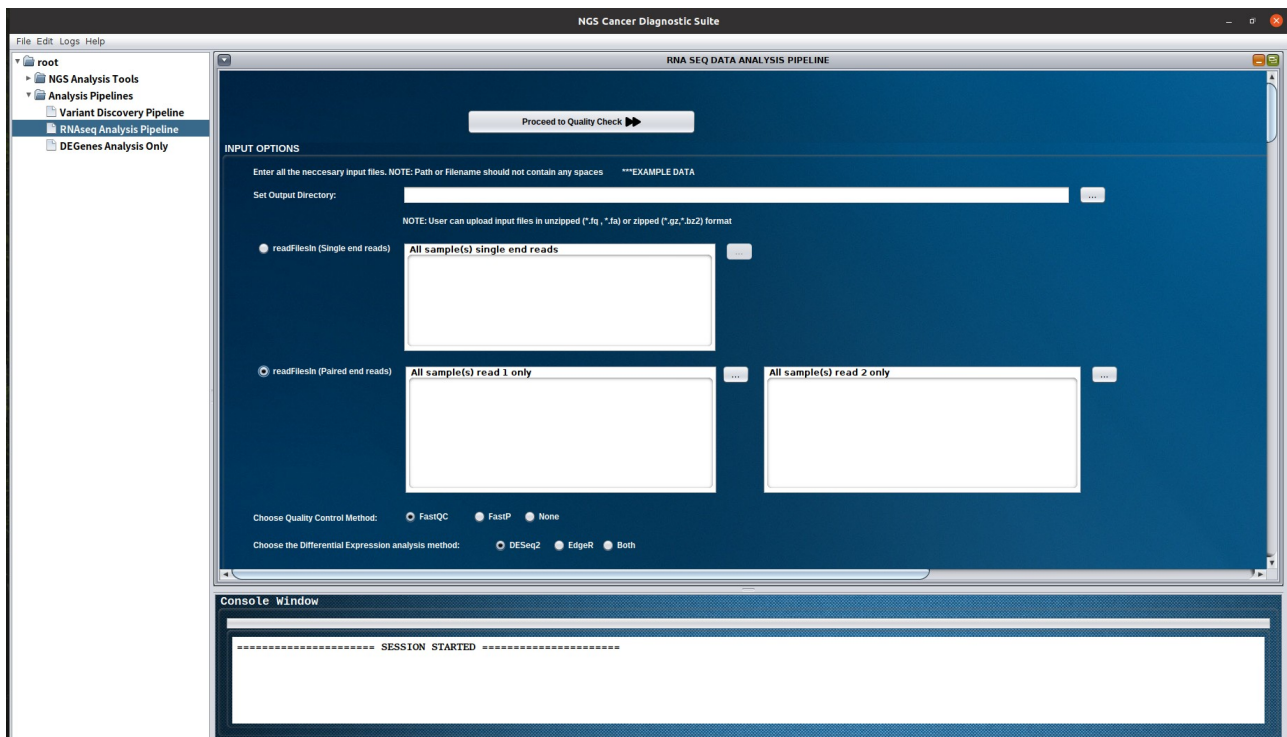


Figure 5: RNA-Seq Pipeline – Main Input Interface

Click the browse “...” button to choose or create an output directory (Figure 5(a)).

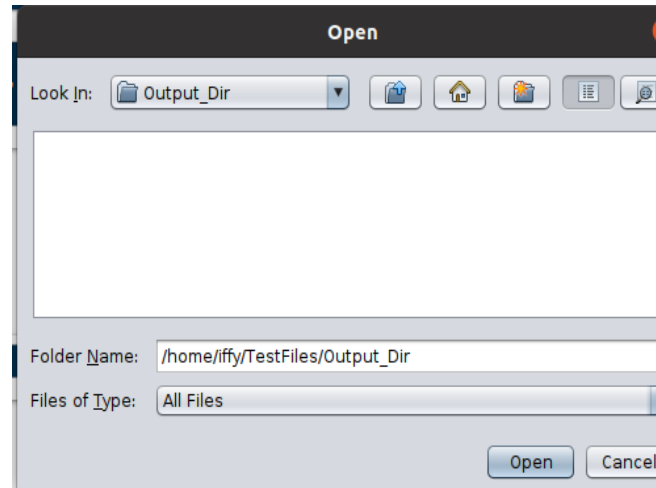


Figure 5(a): Browsing to output directory

Similarly select the input files path by clicking browse “...” button right next to the single-end or paired-end reads. User can upload either all single read samples or paired end read samples in any of the supported format (*.fq, *.fa or *.gz). In case of paried-end reads for ease use the **"Filter: Paired-end Read 1 Files"** option to show only files ending with **_1.fastq.gz** or use the **"Filter: Paired-end Read 2 Files"** option to show only files ending with **_2.fastq.gz** (Figure 5(b)).

Step 2: Quality Check: Read Quality Assessment (Optional)

FastQC with Trimmomatic or FastP (Figure 6(a), 6(b), 6(c)).

- To run trimmomatic and rerun FastQC / To perform additional trimming and rerun FastP
- To proceed to **run pipeline**



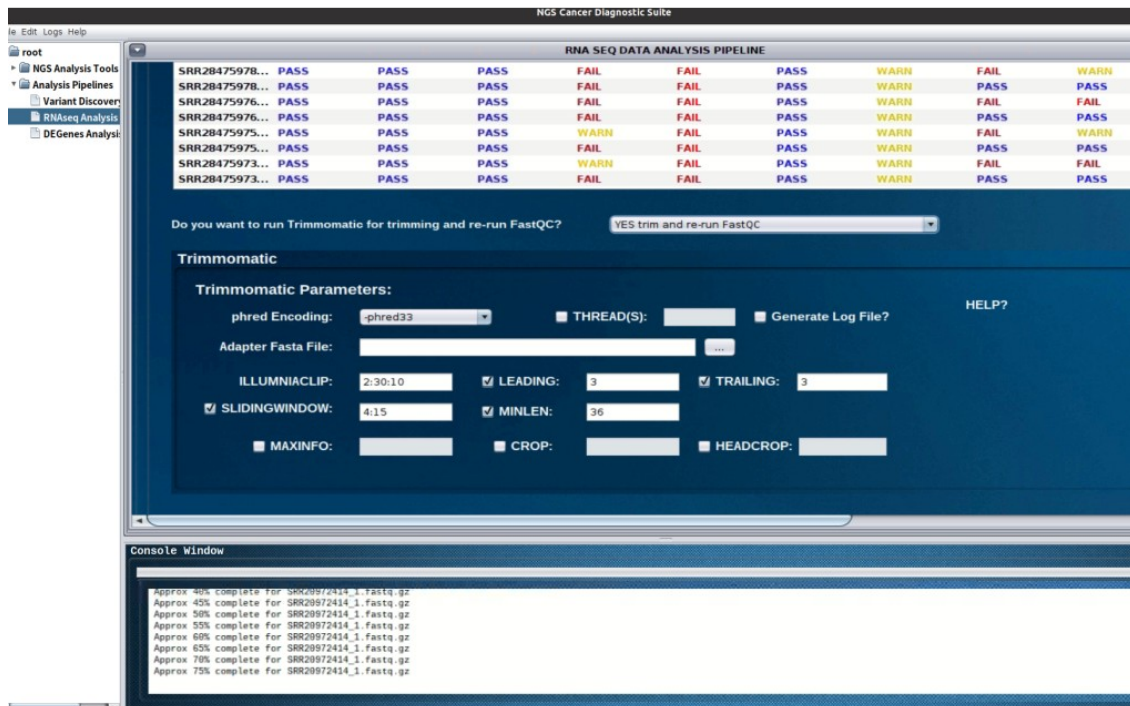


Figure 6(b): Quality Control: FastQC – With Trimmomatic Option Enabled

In 1st case, a folder named **FastQC0** folder will be created. Each time FastQC is re-run after trimmomatic a new folder is created (**FastQC1**, **FastQC2** etc) with updated results. Same is the case with FastP.

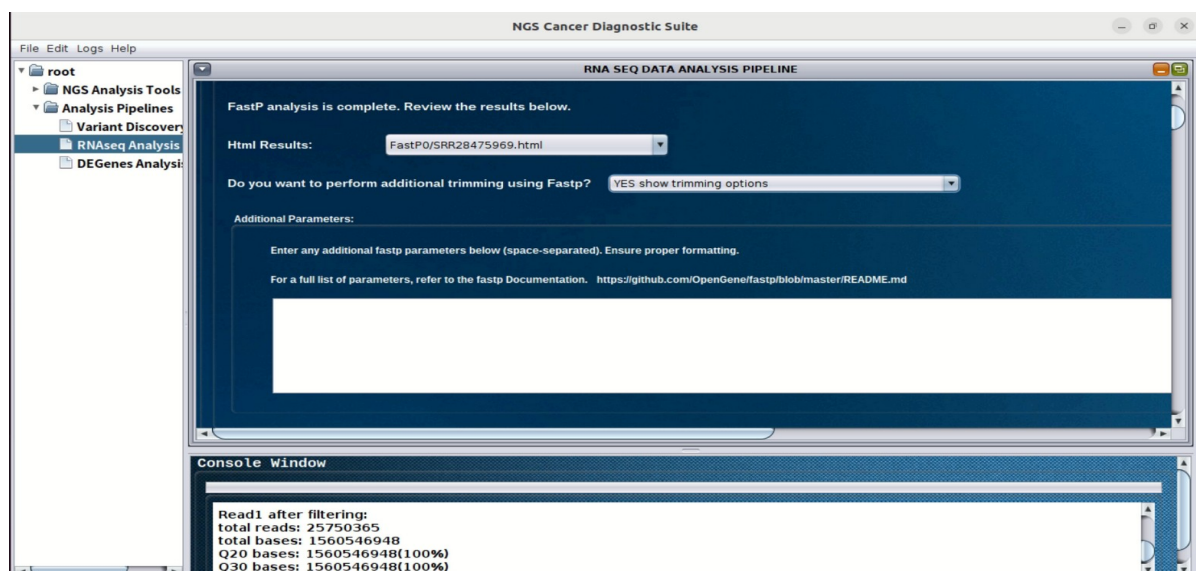


Figure 6(c): Quality Control: FastP – Interface with Adapter Settings

The HTML report by FastQC includes results from multiple modules that were run by FastQC, and provides a quick assessment of the quality of the results labeled as normal (green checkmark), slightly abnormal (orange triangle), and very unusual (red cross) reads. The modules included in the report are as follows:

- **Basic Statistics:** provides introductory compositional statistics such as filename, file type, encoding, total sequences, number of sequences flagged as poor quality, sequence length and %GC for the analyzed read.
- **Per Base Sequence Quality:** displays an overview of the range of quality values across all bases at each position in the FASTQ file. The graph is vertically partitioned into three quality ranges: good (green), reasonable (orange), poor (red).
- **Per Sequence Quality Scores:** displays the quality score distribution over all the sequences, which allows users to see if a subset of the sequences have universally low-quality values.
- **Per Base Sequence Content:** displays the proportion of each DNA base (A, T, C, G) called at a given position in all the sequences.
- **Per Sequence GC content:** displays the GC distribution over all the sequences across the whole length and compares it to a modeled normal distribution of GC content.
- **Per Base N Content:** displays the percentage of base calls at each position for which an N was called. N at a given position indicates the inability to make a normal base call with sufficient confidence.
- **Sequence Length Distribution:** displays the distribution of fragment sizes across all sequences and is highly dependent on the sequencing platform.
- **Sequence Duplication Levels:** counts the degree of duplication for every sequence in a library and creates a plot showing the proportion of sequences with different degrees of duplication (in blue) and de-duplicated sequences (in red).
- **Overrepresented Sequences:** lists all of the sequence which make up more than 0.1% of the total. An overrepresented sequence implies that either it is highly biologically significant, the library is contaminated, or is not that diverse.
- **Adapter Content:** displays the proportion of sequences which have an adapter sequence at a given position. This is informative in deciding whether there is a significant amount of adapter present in the sequences and can be subjected to trimming.
- **Kmer Content:** displays relative K-mer (k=7) enrichment over the read length for the top six K-mers. More specifically, it measures the number of each 7-mer at each position in the library and then uses a binomial test to look for significant deviations from an even coverage at all positions and reports the 7-mers with positionally biased enrichment.

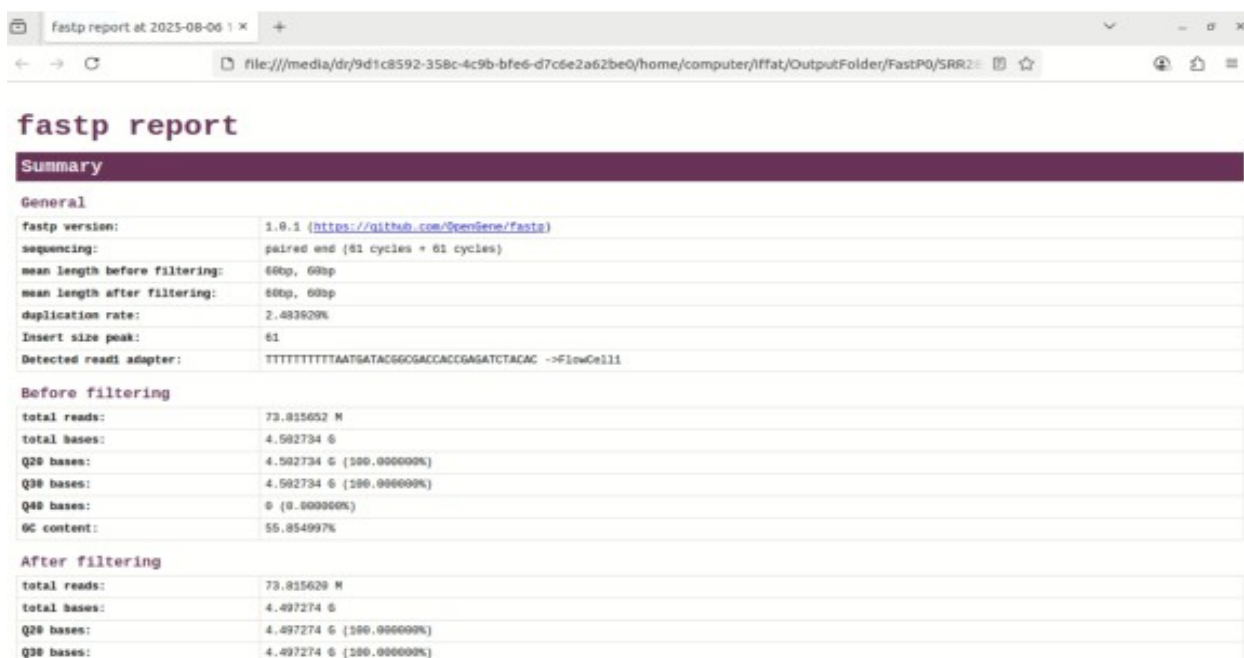


Figure 6(d): Quality Control Reports – HTML Reports Generated by FastQC/FastP

Once the quality control is done user can proceed to **run pipeline**. An other dialog box will appear asking whether to use **original input files** or the **quality-controlled files** for further analysis. After selecting the **input type**, you can press **NEXT** button to view and configure the remaining pipeline settings (Figure 7).

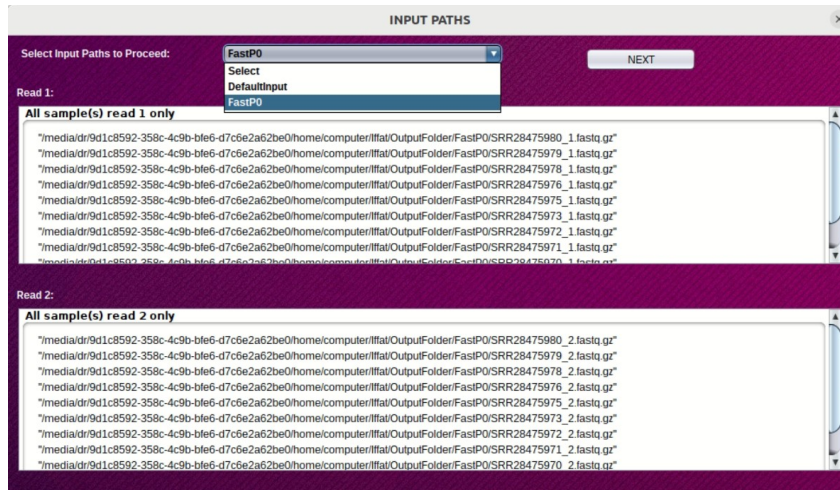


Figure 7: Final Input Dialog – Select Input Files for Processing

Step 3: Metadata Preparation

In next interface the user has to provide or prepare the sample metadata. Two options are available:

- **Upload a sample data csv/tsv file:** containing sample information. Ensure that the first column header is named as “SampleName”.
- **Manual data entry:** Manually add or delete columns and fill in all the rows to create metadata directly within the interface.

Once metadata is finalized, click “**Lock Table**” button to proceed further (Figure 8(a),8(b)).

NGS Cancer Diagnostic Suite

RNA SEQ DATA ANALYSIS PIPELINE

Run Pipeline

BACK

Prepare Sample Meta Data

Please choose one of the following options to provide your sample meta data:

- ☐ Upload a Sample data CSV/TSV File:
Ensure the file has 'SampleName' as the first column (SampleName, condition1, condition2).
- ☒ Manual Data Entry:
Enter data directly into the table. You can add or delete columns as needed to customize your input.

Add Column Delete Column

Done with editing? Lock Table

Press the Lock Table button to proceed with further analysis.

SAMPLENAME
SRR28475969
SRR28475970
SRR28475971
SRR28475972
SRR28475973
SRR28475975
SRR28475976
SRR28475978
SRR28475979
SRR28475980

Console Window

SESSION STARTED

Figure 8(a): Metadata Preparation – Upload or Enter Metadata Table

NGS Cancer Diagnostic Suite

RNA SEQ DATA ANALYSIS PIPELINE

Run Pipeline

BACK

Prepare Sample Meta Data

Please choose one of the following options to provide your sample meta data:

- ☐ Upload a Sample data CSV/TSV File:
Ensure the file has 'SampleName' as the first column (SampleName, condition1, condition2).
- ☒ Manual Data Entry:
Enter data directly into the table. You can add or delete columns as needed to customize your input.

Add Column Delete Column

Done with editing? Lock Table

Press the Lock Table button to proceed with further analysis.

SAMPLENAME	CONDITION	TB_STATUS	HIV_STATUS	GENDER
SRR28475969	control	tb_neg	hiv_neg	male
SRR28475970	control	tb_neg	hiv_neg	female
SRR28475971	control	tb_neg	hiv_neg	female
SRR28475972	tb	tb_pos	hiv_neg	male
SRR28475973	tb	tb_pos	hiv_neg	male
SRR28475975	control	tb_neg	hiv_neg	female
SRR28475976	tbhiv	tb_pos	hiv_pos	male
SRR28475978	tb	tb_pos	hiv_neg	male
SRR28475979	tb	tb_pos	hiv_neg	male
SRR28475980	control	tb_neg	hiv_neg	female

Console Window

SESSION STARTED

Figure 8(b): Metadata Preparation – Locking the Metadata Table

Step 4: Alignment & Quantification

Once the metadata table is locked, scroll down to configure the parameters for **STAR aligner** (Figure 9). For sample mapping user has two options out of which anyone can be selected. Option “Single Sample Per Run” is recommended:

Select Sample Mapping: Single Sample Per Run (Generate separate bam file for each Sample) (Recommended)

Select Sample Mapping: Multiple Reads as Single Sample in Single Run (Generate one bam file for all reads)

Two options are provided for generating **featureCounts** file. You can generate:

- Separate featureCounts file(s) for each sample
- A combined featureCounts(s) file for all samples

The screenshot displays the 'RNA SEQ DATA ANALYSIS PIPELINE' window of the NGS Cancer Diagnostic Suite. The interface is titled 'Provide following information to perform two pass mapping'. It features a sidebar on the left with a tree view containing 'root', 'NGS Analysis Tools', 'Analysis Pipelines', 'Variant Discovery', 'RNAseq Analysis', and 'DEGenes Analysis'. The main panel contains various configuration fields for STAR alignment. On the left, fields include 'No. of Threads' (1), 'Run Mode' (alignReads), 'readFilesCommand' (gunzip -c), 'twopassMode' (Basic), 'outFileNamePrefix' (Combined_Sample), 'sjdbOverhang' (100), 'outSAMtype' (BAM SortedByCoordinate), 'limitBAMsortRAM' (0), 'limitOutSJcollapsed' (1000000), 'quantMode' (GeneCounts), 'outBAMsortingBinsN' (50), and 'outSAMmapqUnique' (255). On the right, 'Genome Directory' and 'Reference Genome' are set to paths under '/media/dr9d1/c8592-358c-4c9b-bfe6-d7c6e2a62be0/home/computer/ffat/Genomes_indexed'. Below these, 'sjdbGTFfile' is set to '/media/dr9d1/c8592-358c-4c9b-bfe6-d7c6e2a62be0/home/computer/ffat/Genomes_indexed/GRCh38.p13.genome.fa'. A section titled 'featureCounts Setting for Multiple Samples' contains two radio buttons: 'Generate separate featureCounts file(s) for each sample' (selected) and 'Generate a combined featureCounts file for all samples'. At the bottom, a 'Console Window' shows the message '***** SESSION STARTED *****'.

Figure 9: STAR Aligner Settings – Configure STAR Alignment Parameters

Step 5: Differential Expression Analysis

Next, configure the settings for **DESeq2** and/or **EdgeR** by selecting the appropriate **analysis type**:

- Single Factor(Simple)
- Multiple Factors(complex)

Set the **design formula** and **contrast groups** for differential expression analysis (Figure 10).

The screenshot displays the 'RNA SEQ DATA ANALYSIS PIPELINE' window within the 'NGS Cancer Diagnostic Suite'. The left sidebar shows a tree view with 'root', 'NGS Analysis Tools', 'Analysis Pipelines', 'Variant Discovery', 'RNAseq Analysis', and 'DEGenes Analysis'. The main panel is titled 'Analysis Mode:' and contains the following sections:

- Please select the appropriate analysis type:**
 - ☐ **Single Factor(Simple)**
For DESeq2, this refers to sample metadata containing one condition column (excluding the SampleName column).
For edgeR, this refers to a pairwise comparison between two groups.
 - ☒ **Multiple Factors(Complex)**
For DESeq2, this applies to sample metadata with more than one condition column.
For edgeR, this involves comparisons with more than two groups or factors.
- edgeR Design**
Create Design Formula:
- Counts Threshold for Filtering:**
- DESeq2 Analysis Parameters:**
 - Search By:**
 - ☒ **By Contrast**
 - Select Factor:
 - Select Factor Level (Reference): VS
 - Select Factor Level: VS
 - ☐ **By List**
 -

At the bottom, a **Console Window** shows the message: '----- SESSION STARTED -----'.

Figure 10: DESeq2 Settings – Differential Expression Analysis Configuration

In case of **edgeR**, select the appropriate statistical test that best suites your analysis (Figure 11).

Following six test combinations are available:

- exactTest,
- glmQLFit+glmQLFTest(coef),
- glmQLFit+glmQLFTest(contrast),
- glmFit+glmLRT(coef),
- glmFit+glmLRT(contrast),
- glmTreat

The screenshot shows the 'RNA SEQ DATA ANALYSIS PIPELINE' window. On the left is a sidebar with a tree view containing 'root', 'NGS Analysis Tools', 'Analysis Pipelines', 'Variant Discovery', 'RNAseq Analysis', and 'DEGenes Analysis'. The main panel is titled 'By List' and contains a 'Select from resultNames list:' dropdown menu with a list of variables including 'Intercept', 'TB_STATUS_tb_pos_vs_tb_neg', 'GENDER_male_vs_female', 'CONDITION_hiv_vs_control', 'HIV_STATUS_hiv_pos_vs_hiv_neg', 'TB_STATUStb_pos.GENDERfemale', and 'TB_STATUStb_pos.CONDITIONhiv'. Below this are input fields for 'alpha p value cutoff:' (0.1), 'padj less than:' (0.05), and 'log2FoldChange greater than:' (2). A button 'Rerun DeSeq Results' is present. The 'edgeR Results' section has a 'Select test:' dropdown, a 'Select two coefficients' section with a list box containing 'hiv', 'tbhiv', 'control', and 'tb', and a 'Contrast:' input field (0.-1.1). It also has 'PValue Threshold:' (0.05) and 'log2FoldChange Threshold:' (2) fields. A 'Generate Results' button is at the bottom right. A 'Console Window' at the bottom shows '***** SESSION STARTED *****'.

Figure 11: edgeR Settings – Differential Expression Analysis Configuration

Step 6: Gene Set Enrichment Analysis

Then configure any parameters for gene set enrichment analysis. Gene Set Enrichment Analysis (GSEA) is a computational method used to analyze high-throughput gene expression data, to identify gene sets or biological pathways that are significantly associated with a particular phenotype or experimental condition (Figure 12(b), 12(b)).

From Type option allows user to chose the format type of genes that is **Ensemble**, **Symbol**, **Uniprot**, **EntrezID**.

Annotation DB organism option requires a valid database provided by bioconductor AnnotationData packages (<https://bioconductor.org/packages/3.22/data/annotation/>).

KEGG Organism requires a valid organism identifier from https://www.genome.jp/kegg/catalog/org_list.html

User can set rest of the parameters accordingly.

Type of analyses included in gene set enrichment analysis are as follows:

- 1 GO Classification (groupGO)
- 2 GO over-representation analysis (enrichGO)
- 3 GO gene set enrichment analysis (gseGO)
- 4 KEGG over-representation analysis (enrichKEGG)
- 5 KEGG pathway gene set enrichment analysis (gseKEGG)
- 6 KEGG module over-representation analysis (enrichMKEGG)
- 7 KEGG module gene set enrichment analysis (gseMKEGG)
- 8 KEGG Structural Pathway Visualization (Pathview)
- 9 Reactome Pathway Profiling
- 10 Human Disease Ontology (DO) Integration

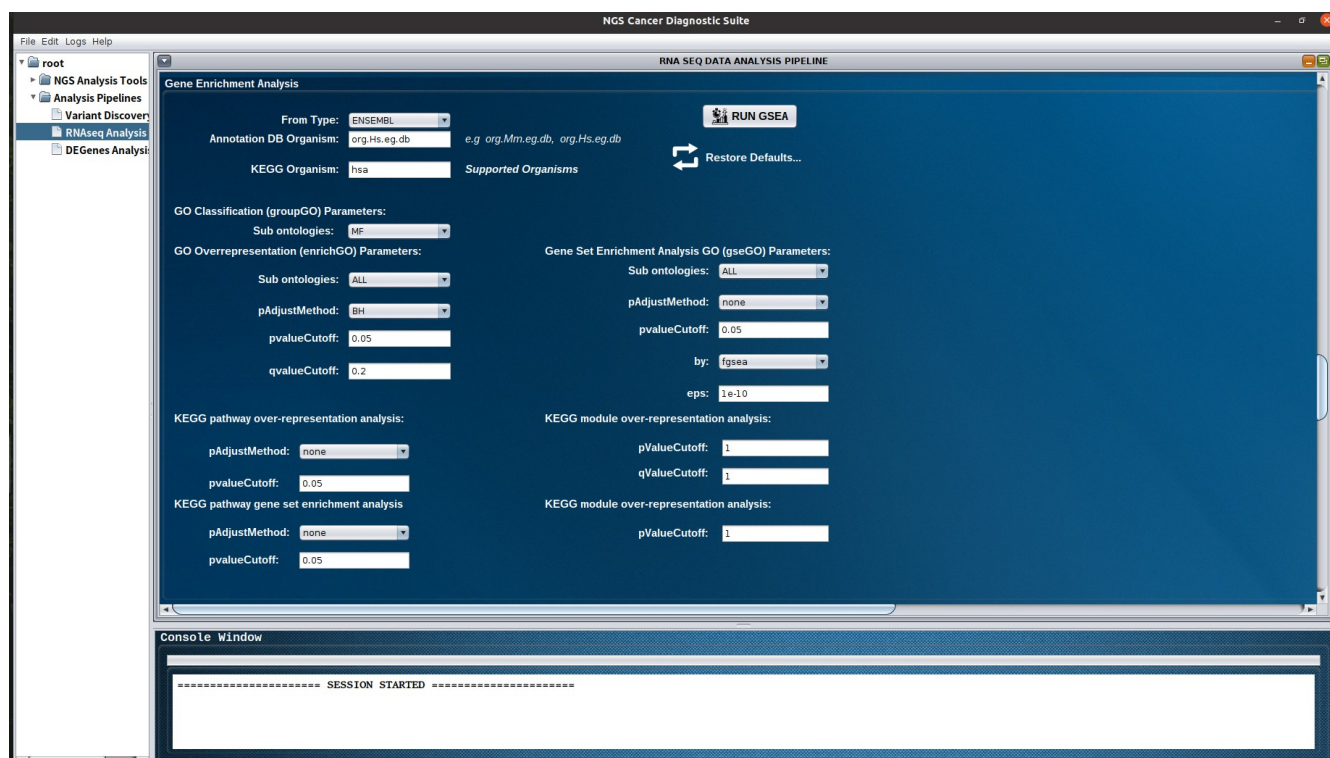


Figure 12(a): Gene Set Enrichment Analysis – Configuration Settings

After setting all the required options in each section, click “**Run Pipeline**” button at the top.

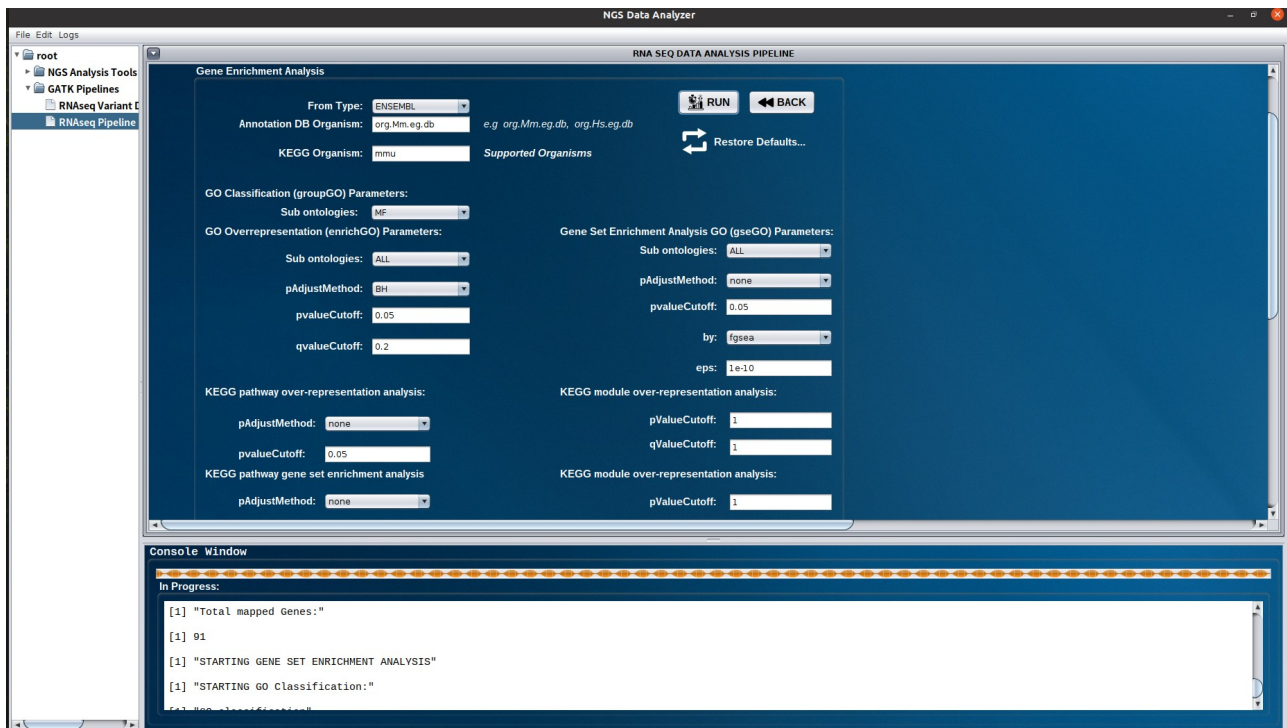


Figure 12(b): Gene Set Enrichment Analysis – Execution in Progress

Once the analysis is completed, user can easily explore the results by exploring folder named **DeSeqResults** or **EdgeRResults** located in output directory (Figure 13(a),(b)). All the graphs and .csv files related to differential expression analysis are stored in these folder.

If you want to re-run **DESeq2** or **EdgeR** analysis with different settings or to correct an error, you can simply click **Rerun Deseq results** or **Generate Results** button.

Important: Before re-running, make sure to **copy** or **rename** the already generated **DeSeqResults** or **EdgeRResults** folders, as they will be **overwritten** with newer results.

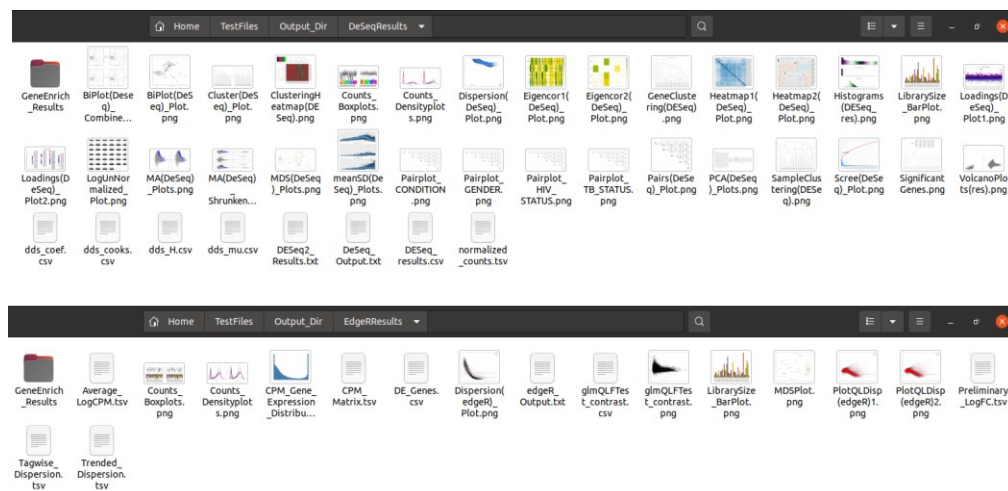


Figure 13(a): DESeqResults / EdgeRResults Folder – Showing Output Files Generated After Differential Expression Analysis

Visualization:

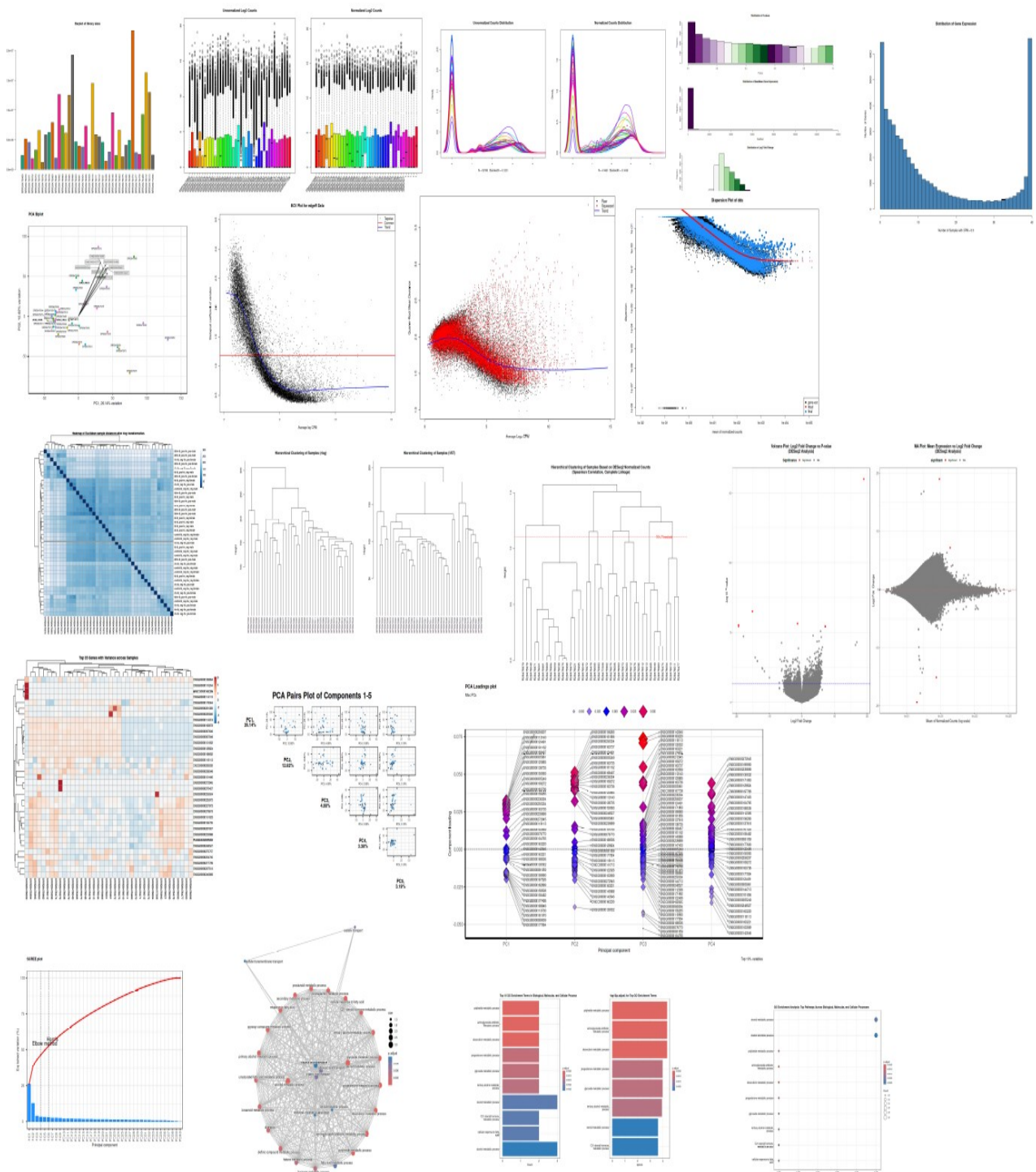


Figure 13(b): Visualization – Displaying Graphical Outputs from Differential Expression Analysis

All logs generated during processing are automatically saved in the **logs** folder located within the software's main directory.

For convenience, users can **right-click** within the **log output area** at the bottom of the interface to save the currently displayed log output as a **.txt** file. This feature is useful for saving session-specific outputs or sharing logs for support and debugging (Figure 14).

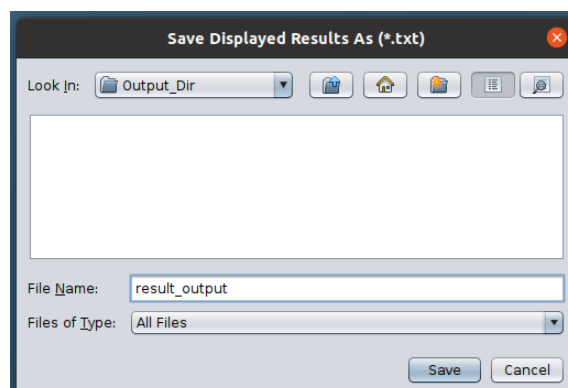


Figure 14: Saving Results from the Log Output Area – Demonstrating Log Export Functionality

Status Tracking Feature:

RNA-seq pipeline also has a status tracking feature. When it is executed from the beginning, a file named **RNASeqPipeline_status.txt** is automatically generated in the output directory (Figure 15). This file tracks the completion status of **each sample's STAR alignment**. If the application is interrupted (e.g., due to a crash or power outage), the pipeline will resume from where it left off when restarted with the same input and output directories. This ensures that already processed samples are not rerun, significantly saving time and computational resources. This mechanism is particularly useful for batch processing and tracking progress over multiple sessions.

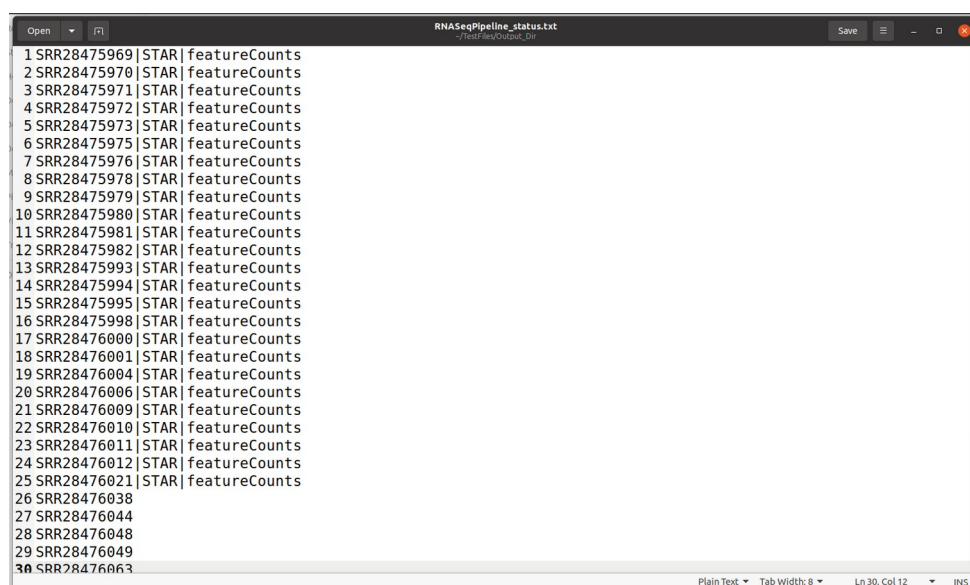


Figure 15: RNA-Seq Pipeline-Status Tracking Feature (RNASeqPipeline_status.txt)

5.2 Differential Expression Gene Analysis Pipeline

Differential gene expression analysis is one of the most common application of RNA-seq data. It refers to the analysis and interpretation of differences in abundance of gene transcripts within a transcriptome. This pipeline interface simplifies the process by allowing users to begin the analysis directly from **counts files**, continuing all the way through **gene set enrichment analysis** (Figure 16).

All configuration steps are the same as those in the RNA-seq analysis pipeline from metadata preparation. The key difference is that this pipeline skips the Quantification and Alignment steps.

At the initial interface, the user is required to provide the following information:

- Set output directory, where the results and intermediate files will be stored.
- Input of counts files, user is provided with two options:
 - User can specify the directory path for the featureCounts folder, where all the count files (*.featureCounts) are present
 - Upload a counts file in *.csv or *.tsv format
- Choose the differential expression analysis method : **DESeq2**, **edgeR** or **Both**

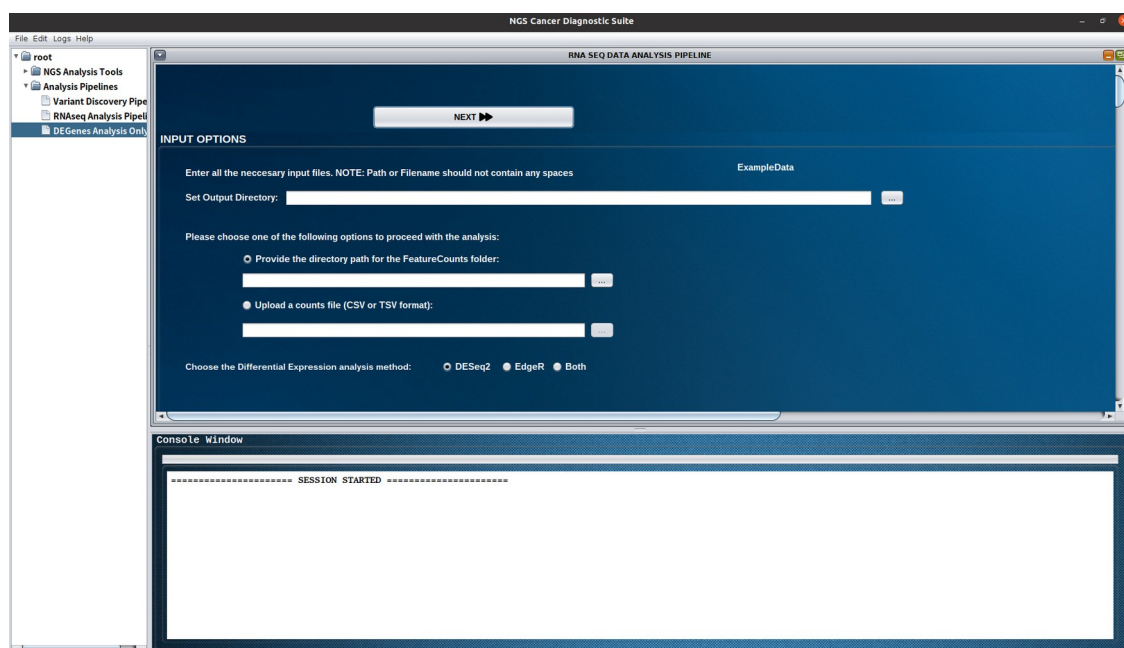


Figure 16: Differential Gene Expression Analysis Interface

5.3 Variant Calling Pipeline

Variant calling involves the identification and characterization of genetic variants such as single nucleotide polymorphisms (SNPs), insertions/deletions (InDels), and structural variants from high-throughput sequencing data. Variant calling is a multi-step process in genomic analysis that involves sequencing, mapping, quantification, calling, filtration and annotation. The **variant calling pipeline** in this suite is designed to automate this end-to-end process, starting from **fastq files** or **aligned bam files** to producing annotated variants reports.

This pipeline includes the following key steps:

- **Pre-processing of FASTQ files:**
This includes quality control and adaptor trimming (if applicable). The user can specify to perform quality check using **FastQC** , **FastP** or **None**
- **Alignment:** Reads are aligned to reference genome using BWA
- **Add or Replace read groups:** BAM files are checked for the presence of read group header. If missing, it's added by **AddOrReplaceReadGroups**
- **Marking Duplicates:** Duplicate reads are identified and marked by **MarkDuplicates**
- **Splitting Cigar Reads (Optional):** splits reads with complex CIGAR strings into smaller, more manageable reads for variant calling. **SplitNCigarReads** is used when required, typically for RNA-seq data.
- **Base Quality Score Recalibration:** Recalibration is performed using **BaseRecalibrator** and **ApplyBQSR**
- **Covariate Analysis:** **AnalyzeCovariates** is then used to generate pdf and csv reports of base quality score recalibration
- **Variant Calling:** Both germline and somatic variant option are available. User can opt for germline variant calling through **HaplotypeCaller** or somatic variant calling through **Mutect2** or even **both**. Notably, the key strength of variant calling pipeline lies in its ability to accommodate both germline and somatic variant calling simultaneously.
- **Variant Selection:** is performed using **SelectVariants** to isolate desired variant types (e.g. SNP, INDELs)
- **Variant Filtration (Pre-annotation):** Filtering is applied using **VariantFiltration** to remove low-quality or undesired variants before annotation.
- **Variant Annotation:** User can specify **snpEff**, **VEP** or **both** to perform annotation
- **Variant Filtration (Post-annotation):** Additional filterig can be done using **snpSIFT**, typically based on annotation fields.

Step 1: Setting up Output Directory and Input files:

This interface allows users to configure the initial settings for running Variant calling pipeline. User can specify:

- **Output directory,**
- **Quality Control Method:** Choose the quality control tool from the options: **FastQC**, **FastP** or **None**

If **FastQC** or **FastP** is selected, click **Proceed to Quality Check** button ,

If **None** is selected then click **NEXT** button.

Note: *Quality control settings are the same as described in the RNA-seq pipeline section above.*

- **Input Options:** Two input options are supported:
 - 1- **FASTQ Files:** Provide raw sequencing reads (single-end or paired-end) to run the full pipeline from the beginning, including alignment using **BWA**.
 - 2- **Pre-aligned BAM Files:** Provide BAM files that are already aligned. In this case, the pipeline skips the alignment step and proceeds with downstream processing.

User can set up any folder as output directory by clicking and browsing to that folder. All the generated results will be stored in that output directory. Its recommended that folder is empty otherwise files might be overwritten (Figure 17(a),(b)).

NOTE: Path should not contain any spaces.

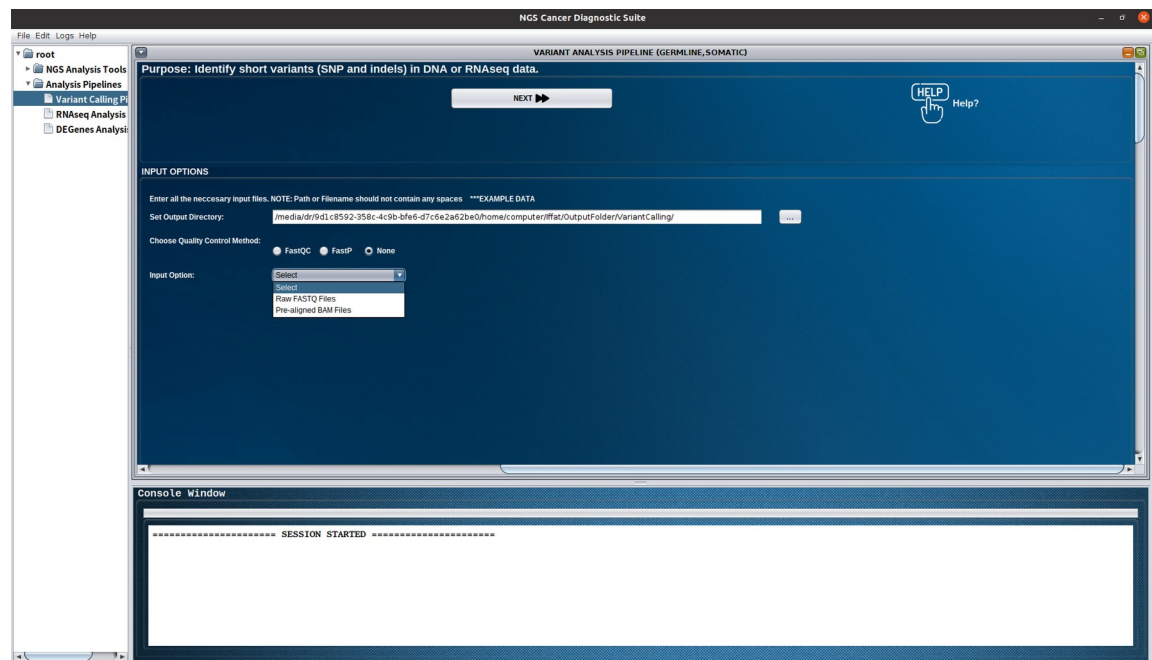


Figure 17(a): Variant Calling – Input Interface

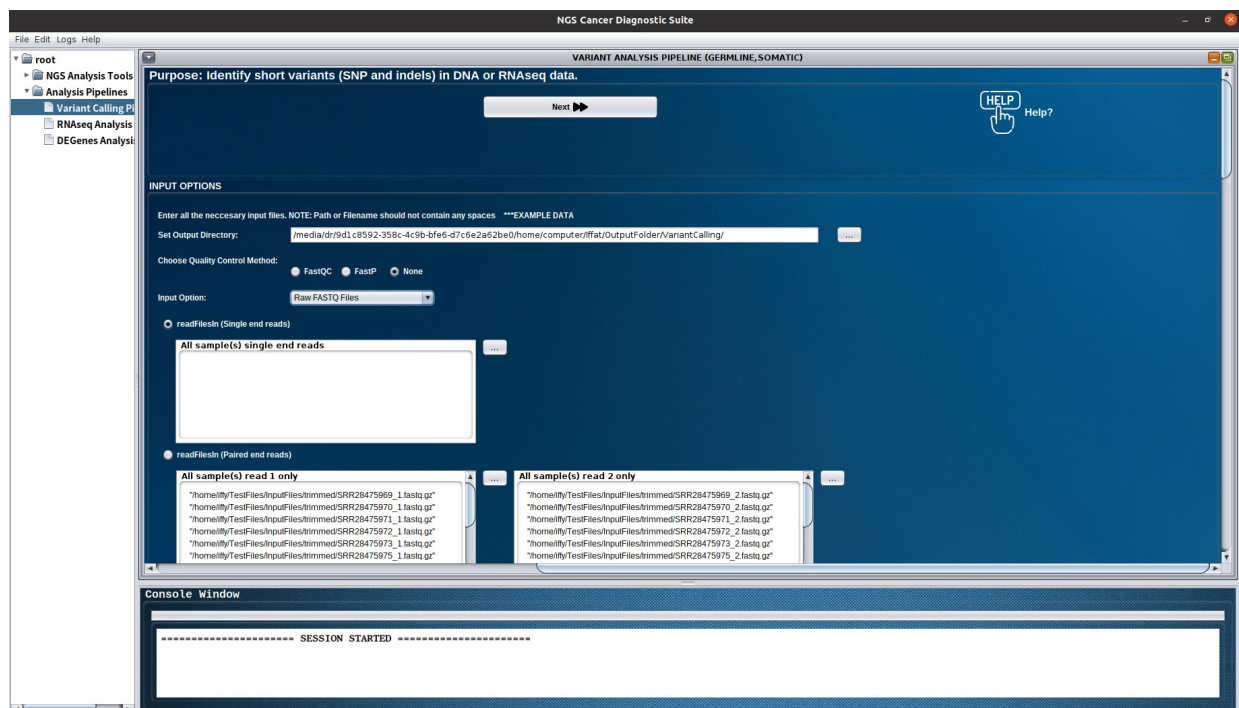


Figure 17(b): Variant Calling – Input File Options

Step 2: Reference Genome, Alignment and Base quality recalibration

In this interface, the user can configure all the required settings for alignment, read group assignment and base quality recalibration (Figure 18).

Reference Genome: Upload the reference genome **FASTA** file to be used throughout the pipeline.

BWA Alignment Settings: Specify the number of **threads**. Optionally enable **-M** argument. The **-M** flag tells BWA to mark shorter split-read alignments as secondary alignments, which is required by some downstream tools (e.g., Picard and GATK).

Base Quality Recalibration Settings: You can upload **.vcf** or **.vcf.gz** for **knownsites** input for **BaseRecalibrator**(e.g., dbSNP or known INDELs).

All other parameters for **BWA**, **BaseRecalibrator**, **ApplyBQSR** will use default settings.

The screenshot displays the 'NGS Cancer Diagnostic Suite' window, specifically the 'VARIANT ANALYSIS PIPELINE (GERMLINE, SOMATIC)' section. The interface is designed for configuring the pipeline settings for identifying short variants (SNP and indels) in DNA or RNAseq data. A 'RUN PIPELINE' button is prominently displayed at the top right of the main configuration area. Below this, the 'Pipeline Settings' section is divided into several input fields and a table. The 'Reference Genome(fasta):' field contains a file path. The 'BWA Options:' section includes a 'Thread(s):' input set to 4 and a checkbox for the '-M' option. The 'AddOrReplaceReadGroups:' section features a table with three rows, each representing a read group with fields for ID, LB, PL, PU, and SM. The 'Base Quality Recalibration:' section includes a text area for 'known Sites (dbSNP):' with several file paths listed. A 'Console Window' at the bottom shows the message 'SESSION STARTED'.

ID	LB	PL	PU	SM
1	lib1	ILLUMINA	unit1	SRR28475969
2	lib2	ILLUMINA	unit2	SRR28475970
3	lib3	ILLUMINA	unit3	SRR28475971

Figure 18: Reference Genome, BWA Alignment, and BaseRecalibrator Settings

Step 3: Splitting CIGAR reads and Variant Calling:

Its optional for the user to select **SplitNCigarReads**, and is primarily used for **RNA-seq data**, where spliced alignments are common (Figure 19, Figure 20(a),(b),(c)).

Variant Calling Configuration: Users can choose between **Germline**, **Somatic**, or **Both** types of variant calling:

1- Germline Variant Calling (HaplotypeCaller): User can select one of the following modes:

- **Single Sample Mode:** Each sample is processed individually. Separate .vcf files will be generated for each sample.
- **Joint Genotyping Mode:** Individual GVCF files are generated using HaplotypeCaller -ERC GVCF, then combined using **GenotypeGVCFs**, resulting in a joint file named **joint_hap.vcf.gz**.
- **Additional Options:** Users can set the **-ERC** (Emit Reference Confidence) mode. **Annotation groups** can be **added** or **removed** using the + and – buttons. Optionally, users can enable **bamout** output generation.

2- Somatic Variant Calling (Mutect2): Users can choose from the following analysis modes:

- Tumor with Matched Normal (Single Pair)
- Tumor with Matched Normal (Multiple Samples)
- Tumor-Only Mode
- Mitochondrial Mode
- Force Calling Mode

Additional Mutect2 parameters can be customized as needed to fine-tune the analysis.

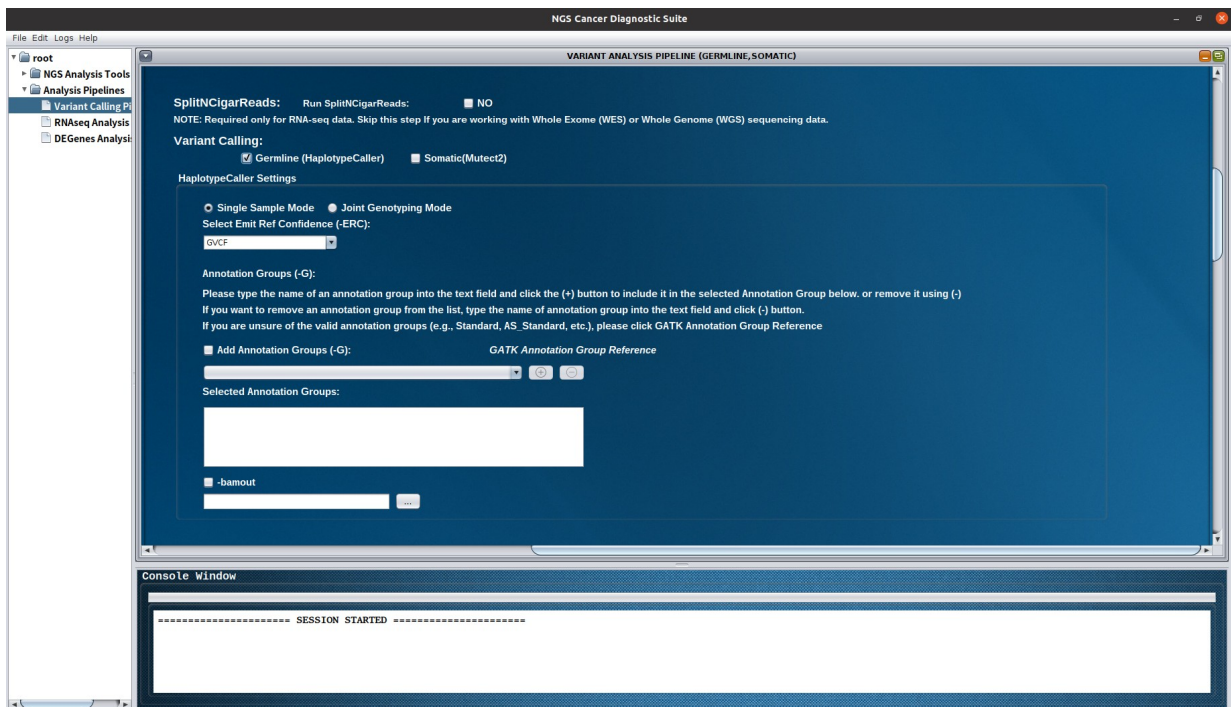


Figure 19: SplitNCigarReads and Germline Variant Calling Settings

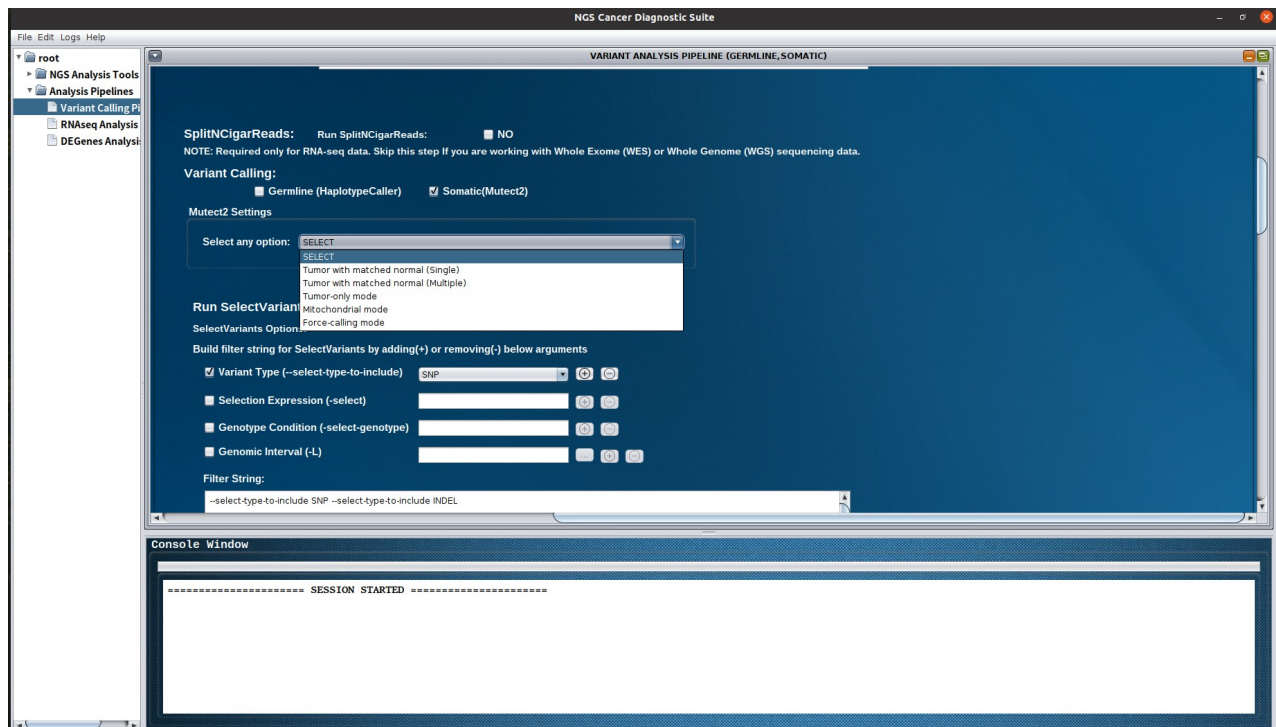


Figure 20(a): Somatic Variant Calling – Mutect2 Selected Mode View

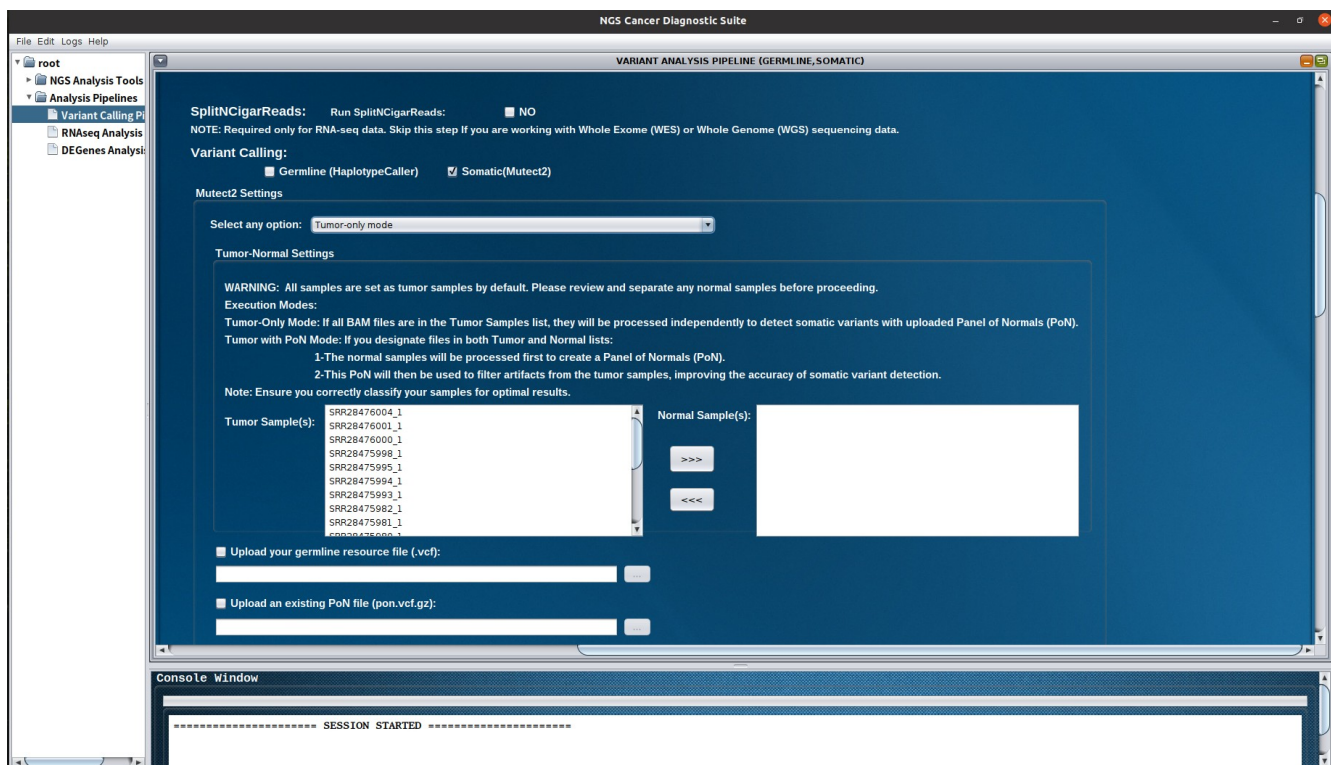


Figure 20(b): Somatic Variant Calling – Mutect2 Parameter Configuration I

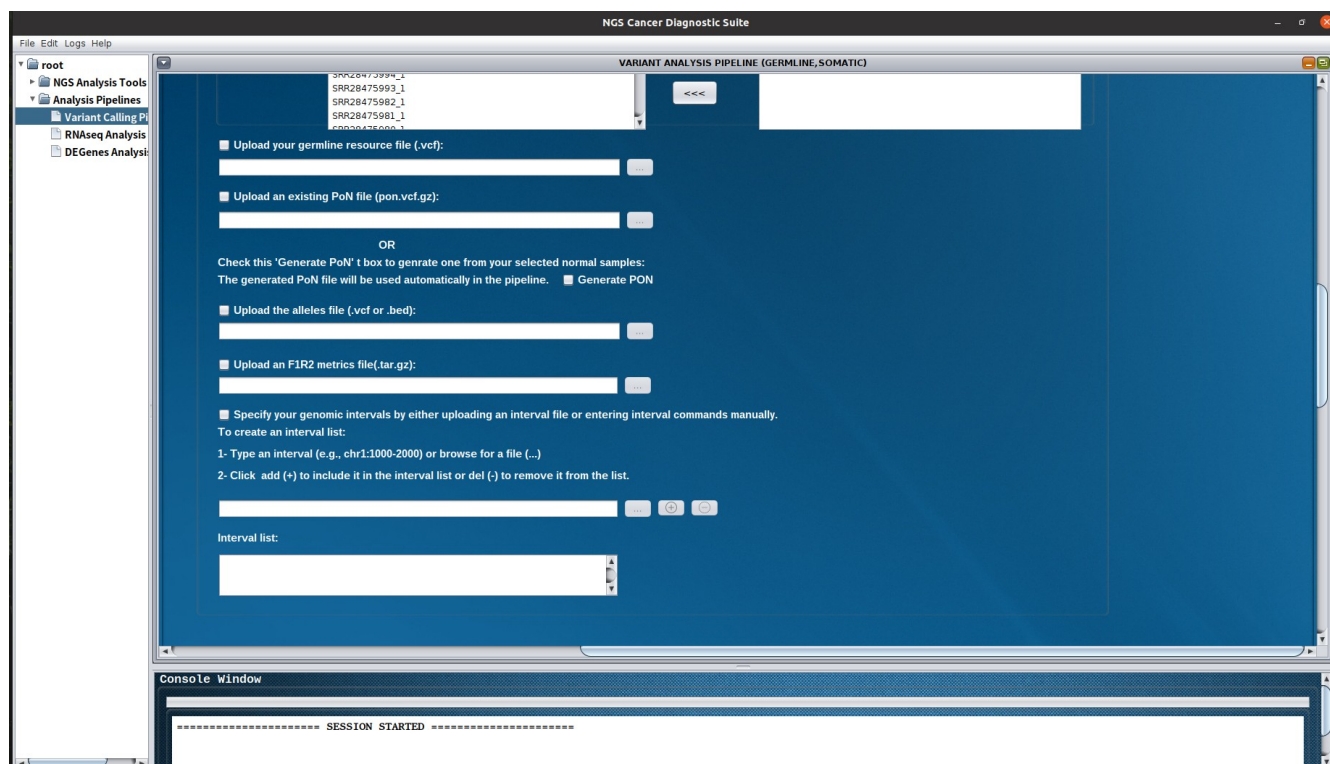


Figure 20(c): Somatic Variant Calling – Mutect2 Parameter Configuration II

Step 4: Variant Filtration (Pre- annotation):

Before annotation, the user must apply at least one filtering step to ensure variant quality. User can chose between **SelectVariants** or **VariantFiltration**. Each option allows the user to configure parameters according to their specific requirements. It is **mandatory** to select at least one of the filtering options before proceeding to annotation (Figure 21).

Note: Proper variant filtering improves the accuracy of downstream annotation and interpretation.

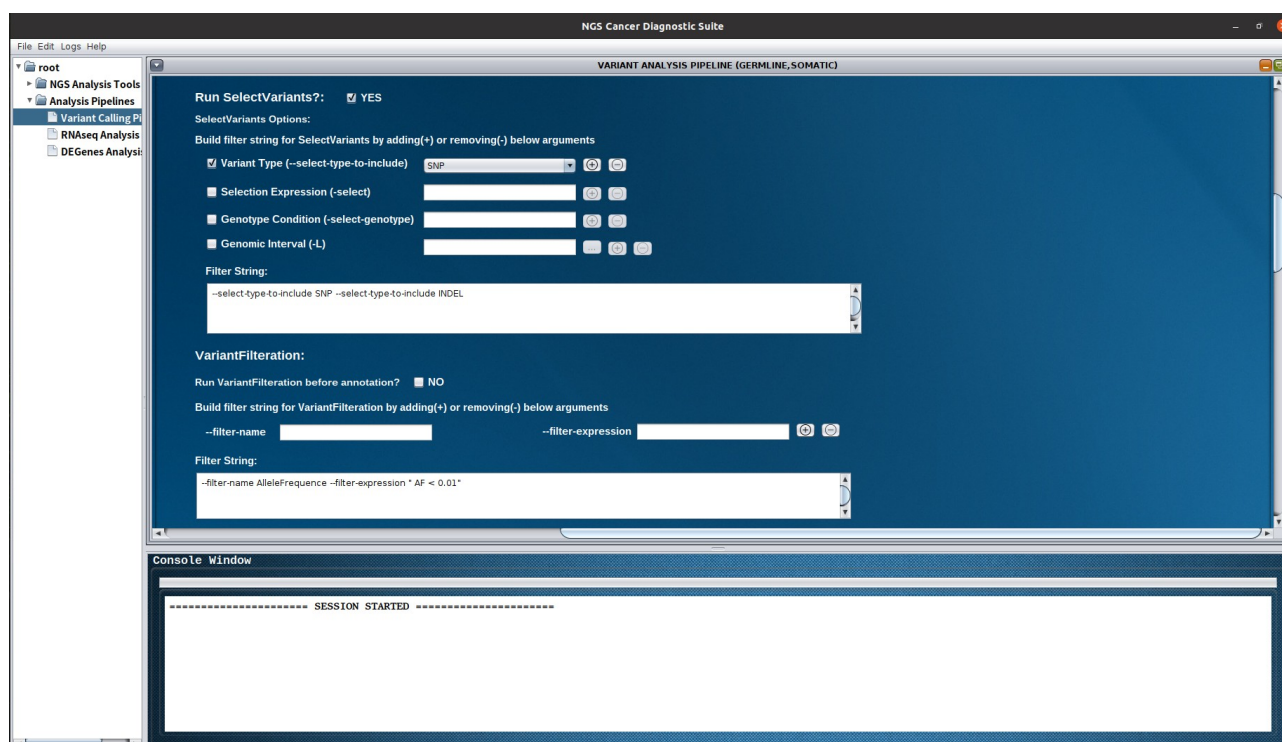


Figure 21: Variant Filtering (Pre-Annotation) – SelectVariants / VariantFiltration Configuration

Step 5: Variant Annotation and Variant Filtration (Post- annotation):

In this step, users can annotate the filtered variants using one or both of the following tools: **snpEff**, **Vep** or **Both**. After annotation, users can optionally apply post-annotation filtering using SnpSift. Filtering parameters can be customized to refine the annotated variants based on user-defined criteria (Figure 22(a),(b)).

Tip: Post-annotation filtering helps in narrowing down clinically or biologically relevant variants by focusing on effect types, impact levels, or custom logic.

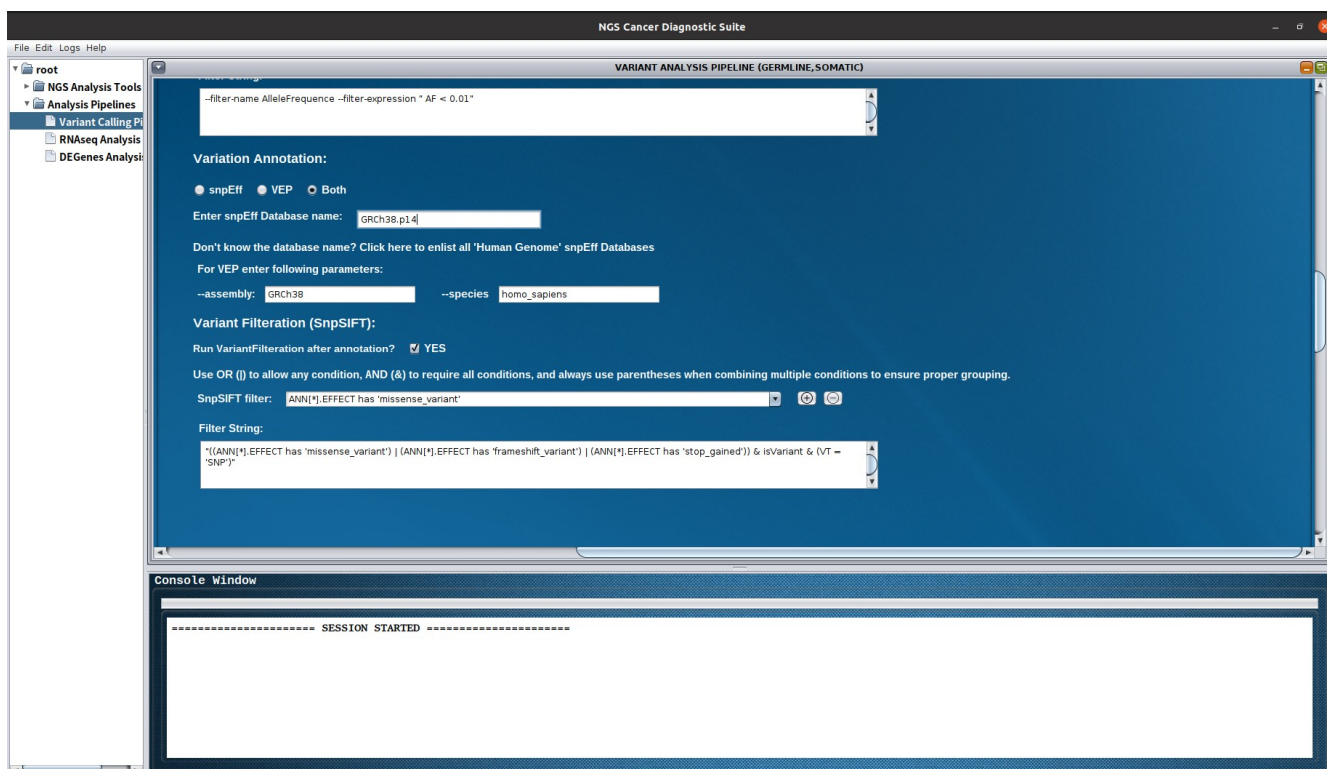


Figure 22(a): Variant Annotation Configuration

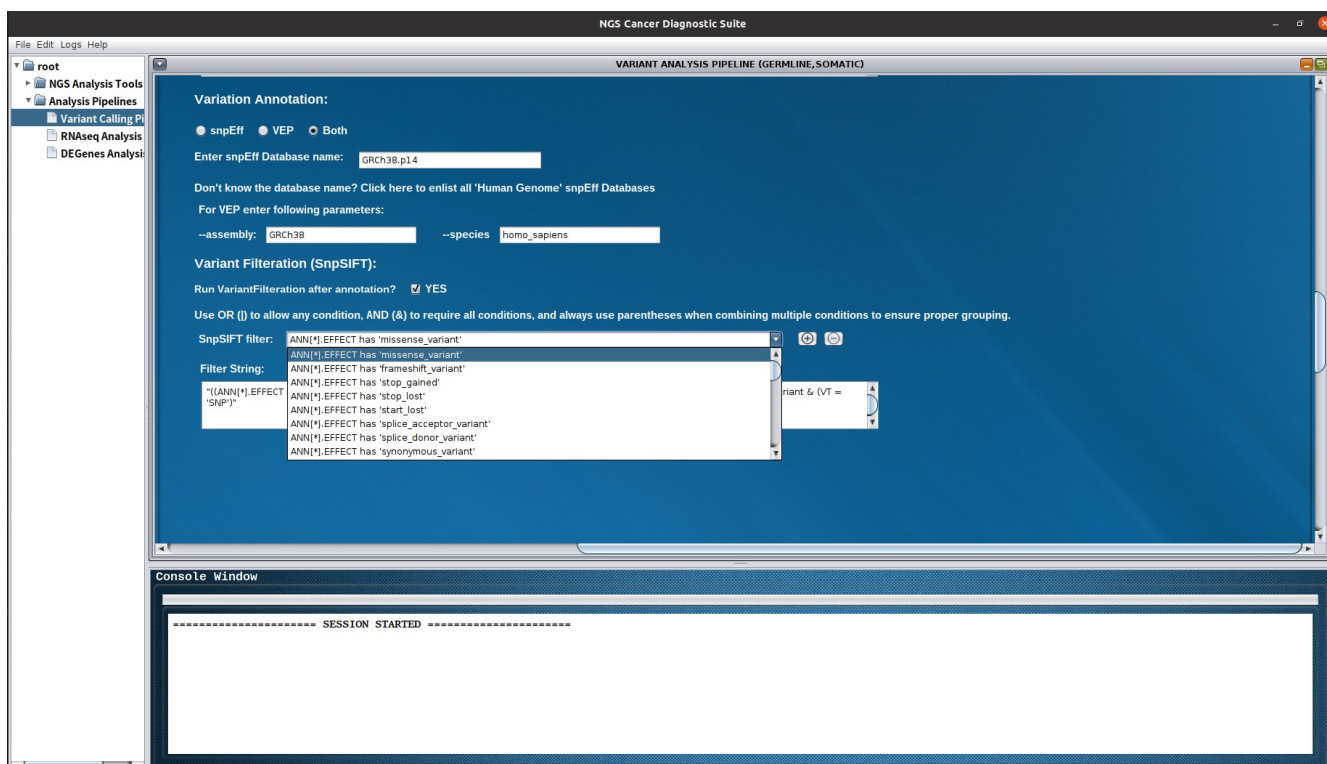


Figure 22(b): Variant Annotation – Post-Annotation Filtering Options

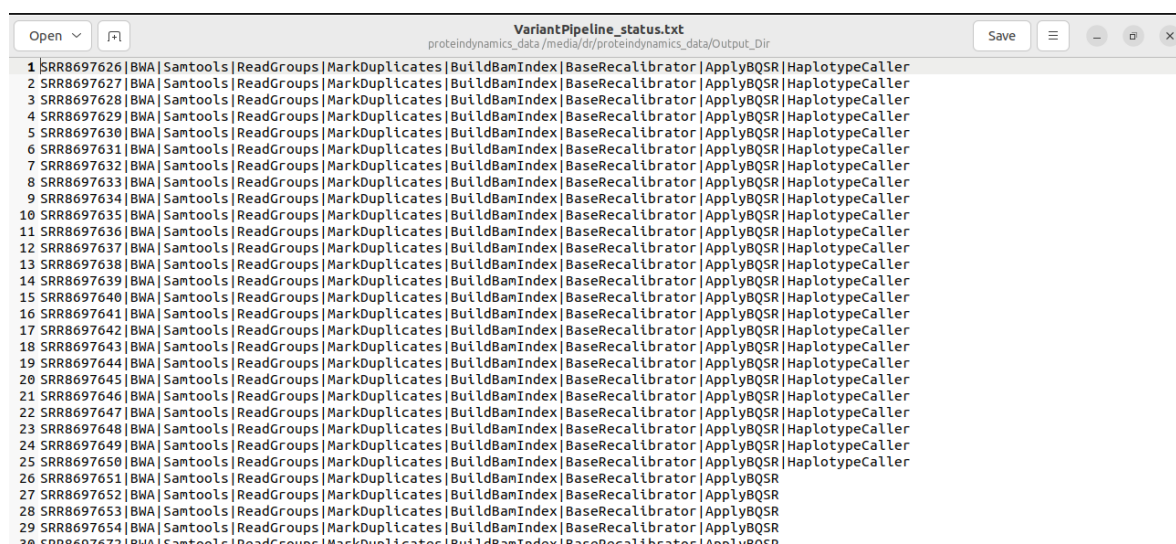
Status Tracking Feature

The Variant Calling pipeline in NGS-CDS is equipped with an automatic **status tracking feature** designed to enhance robustness and ensure seamless recovery in case of interruptions.

A file named **VariantPipeline_status.txt** is generated in the specified **output directory**. This file keeps track of the progress for each sample processed through the pipeline (Figure 23).

If the application is closed unexpectedly—due to power failure, system crash, or manual interruption—the pipeline will **automatically resume from the last completed step** the next time it is run with the **same input and output directories**. It reads the VariantPipeline_status.txt file to determine which samples have already been processed and skips them, preventing redundant computation and saving valuable time.

Additionally, to manage **disk space efficiently**, the pipeline is designed to **automatically remove intermediate files up to the Base Quality Score Recalibration step (BaseRecalibrator)**. All important output files generated **after BaseRecalibration**, including variant calling results, are retained for downstream analysis and traceability.



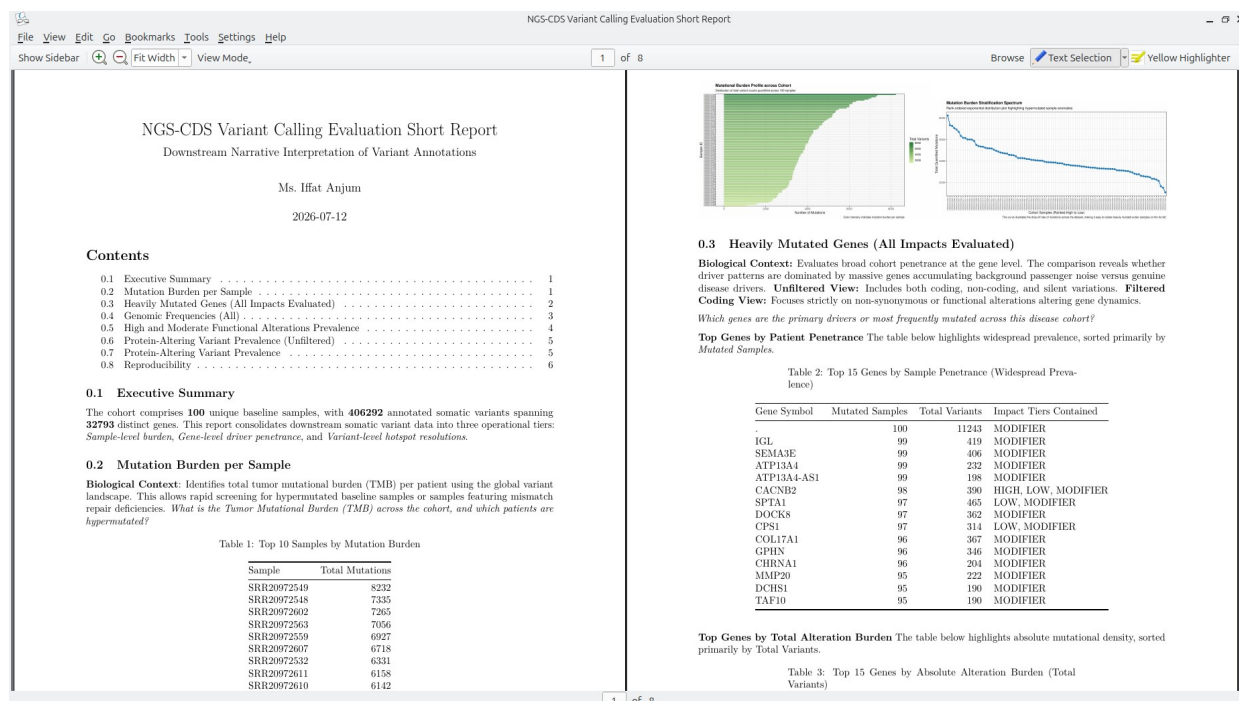
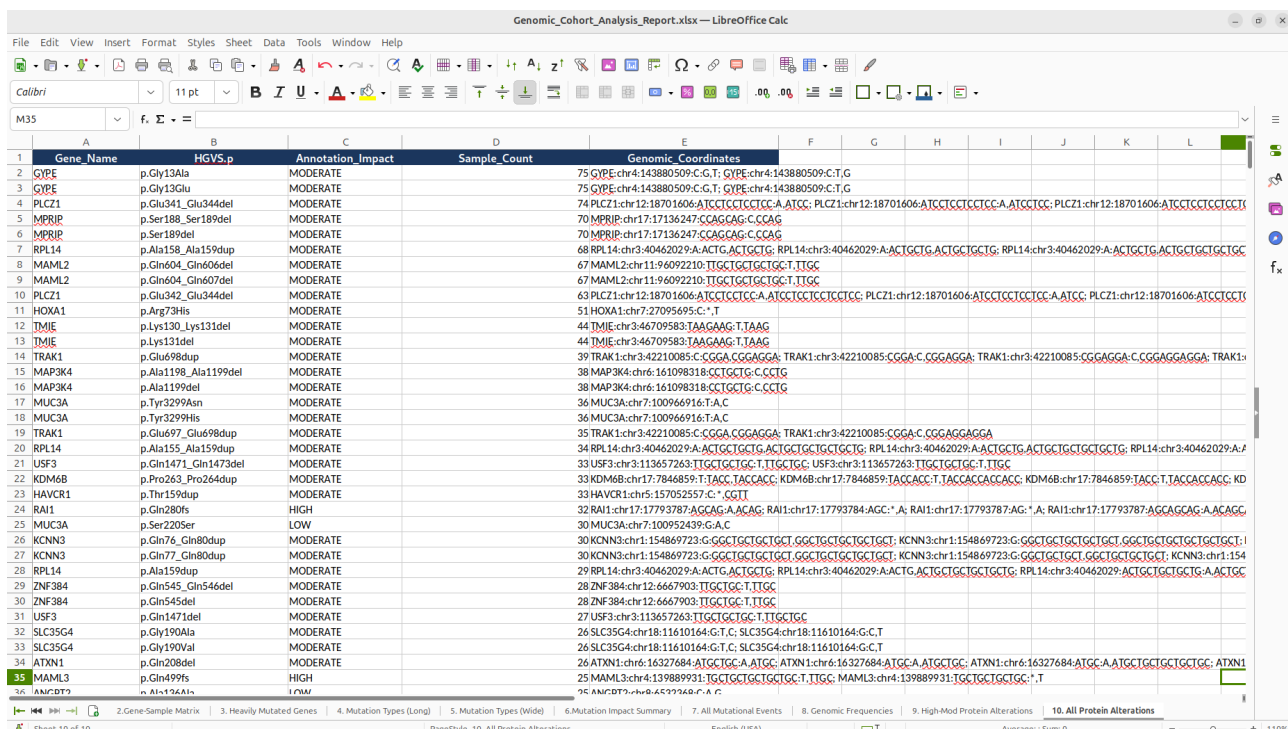
The screenshot shows a text editor window titled "VariantPipeline_status.txt" with a file path "proteindynamics_data/media/dr/proteindynamics_data/Output_Dir". The file contains a table with 30 rows, each representing a sample. The columns are: Sample ID, BWA, Samtools, ReadGroups, MarkDuplicates, BuildBamIndex, BaseRecalibrator, ApplyBQSR, and HaplotypeCaller. The first 29 rows show the first seven steps completed, while the last row (30) shows the first six steps completed. The last row is highlighted in blue.

Sample ID	BWA	Samtools	ReadGroups	MarkDuplicates	BuildBamIndex	BaseRecalibrator	ApplyBQSR	HaplotypeCaller
1 SRR8697626	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
2 SRR8697627	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
3 SRR8697628	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
4 SRR8697629	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
5 SRR8697630	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
6 SRR8697631	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
7 SRR8697632	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
8 SRR8697633	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
9 SRR8697634	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
10 SRR8697635	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
11 SRR8697636	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
12 SRR8697637	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
13 SRR8697638	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
14 SRR8697639	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
15 SRR8697640	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
16 SRR8697641	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
17 SRR8697642	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
18 SRR8697643	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
19 SRR8697644	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
20 SRR8697645	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
21 SRR8697646	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
22 SRR8697647	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
23 SRR8697648	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
24 SRR8697649	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
25 SRR8697650	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
26 SRR8697651	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
27 SRR8697652	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
28 SRR8697653	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
29 SRR8697654	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]
30 SRR8697655	[x]	[x]	[x]	[x]	[x]	[x]	[x]	[x]

Figure 23: Variant Calling Pipeline- Status tracking (VariantPipeline_status.txt)

Once the variant calling analysis is complete, user can review results in the following generated reports:

- **Genomic_Cohort_Analysis_Report.xlsx**: Comprehensive spreadsheet of parsed VCF records, systematically categorized and annotated for easy filtering (Figure 24(a)).
- **Genomic_Cohort_Summary.pdf**: Summary report highlighting key findings, metrics, and visual insights derived from the cohort dataset (Figure 24(b)).
- **Plots/** Directory containing high-resolution, detailed visualizations and individual charts generated during the cohort analysis (Figure 24(c)).



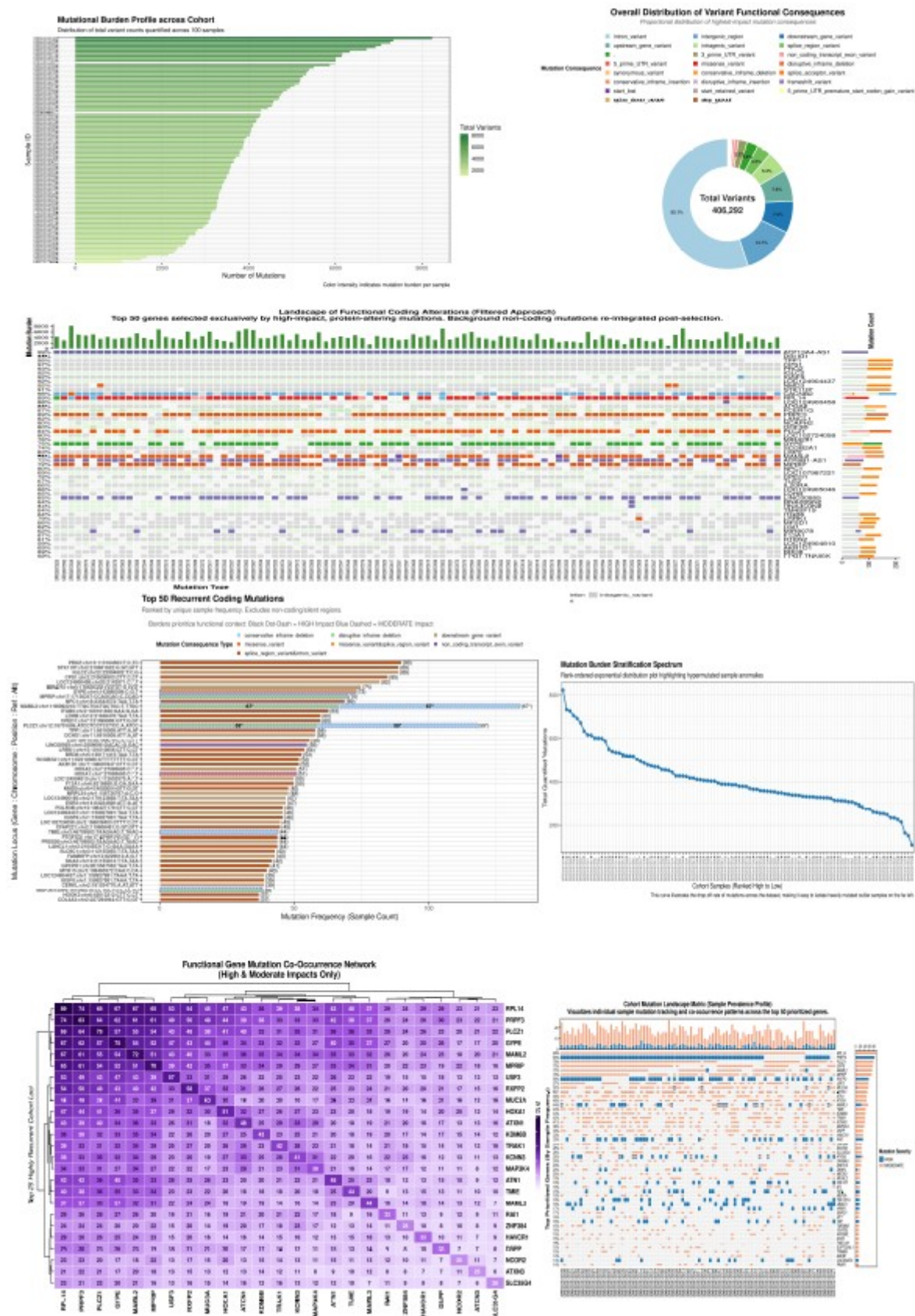


Figure 24(c): Collection of high-resolution cohort visualizations, including oncoprint profiles, tumor mutation burden distributions, gene prevalence rates etc